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Review

Transition metal nanoparticles as nanocatalysts for Suzuki, Heck and Sonogashira cross-coupling reactions

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ABSTRACT

Transition metal (TM) catalyzed cross-coupling reactions are the utmost versatile and reliable methods for the production of many industrially important fine chemicals. The utilization of transition metals nanocatalysts in organic transformations, specifically the Suzuki, Heck, and Sonogashira cross-coupling reactions, have enticed the vigilant eye of researchers and technologists due to; (i) high surface area, (ii) higher number of surface atoms leading to large number of catalytic active centers, and (iii) chemical stability. Moreover, being colloidal, nanomaterials combine the properties of both homogeneous and heterogeneous, i.e., solubility in different solvents and ease in the separation of catalysts. This improves the overall catalytic efficiency and cost. Therefore, these pseudo-homogeneous nanocatalysts are gaining a great deal of attention due to their broad substrate scope and ease of handling. This review article focuses on the recent advances in developing nanostructured catalysts and their utilization for selective carbon–carbon bond-forming reactions, e.g., Suzuki, Heck, and Sonogashira coupling reactions. The main focus is laid on the reactions catalyzed by nanocatalysts derived from transition metals such as Pd, Pt, Cu, Ag, Ni, Fe, Au, and Zn, expounding the conception and rationale based on the reactivity improvement and the recyclability of the nanocatalysts are presented here. The heterogeneous catalysis, essential in daily chemical transformations, is demonstrated, implicating significant functions presented by the nanocatalysts, related pros and cons, recyclability, and efficiency.

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1. Introduction

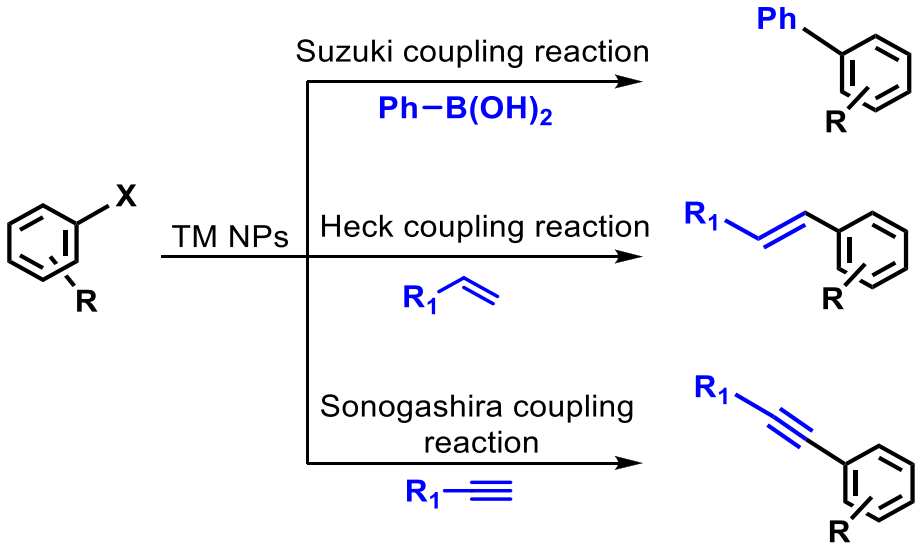
TM-catalyzed cross-coupling reactions are amongst the most powerful tools that find widespread applications in synthesizing natural products, functional materials, pharmaceuticals, and agrochemicals [1,2]. Among them, Suzuki-Miyaura (Suzuki), Mizoroki-Heck (Heck), and Sonogashira cross-coupling reactions are considered the most versatile and broadly applied reactions for the formation of C—C, C═C, and C≡C, respectively [3,4] (Scheme 1).

These three reactions have significant practical and theoretical significance in pharmaceuticals, polymers, and natural product synthesis. These reactions are also used for the intra-cellular synthesis of therapeutic molecules [5]. Traditionally, these reactions are catalyzed by homogeneous Pd-complexed catalysts and ligands such as phosphor and *N*-heterocyclic carbene [6–8]. Despite significant advancements in attaining higher activity, enhanced selectivity, and broader substrate scope, homogeneous Pd catalysts suffer from liabilities such as toxicity and post-processing procurement of most of the ligands used, expensive downstream separation [9] of residual metal from the final product below the maximum allowable level, and reusability of the catalyst [10]. Moreover, the use of toxic solvents, cumbersome multi-step synthesis involved in their preparation, tedious handling due to their air sensitivity, and laborious chromatographic purification of the product also pose a challenge. Consequently, homogeneous Pd catalysts are far from industrial realism [11,12].

Recently, heterogeneous metal catalysis has been gaining much attention due to its superior catalytic activity, broad substrate scope, and ease of handling [13]. The nanocatalysts are easy to recover by filtration or centrifugation, allowing reusability of the catalyst after the completion of the reactions [3,14,15]. A great part of the former heterogenous catalysis community is emerging into the nanoparticle community, as highlighted in the special issue of the prestigious journal "*Chem. Rev.*"(nanocatalysts) [16]. Moreover, nanocatalysts-based transformation is considered a non-hazardous and greener approach, which is desirable for sustainability and economic viability [17]. In addition, nanocatalysts that are built up on TM nanoparticles (NPs) such as Au, Cu, Pd, and their alloys can drive a wide range of C—C cross-coupling reactions and C-Het (N, S, N) bond formations with enhanced catalytic activity, broad substrate scope, and improved yields [15]. These catalyst systems are generally free from ligands and the requirement of air and moisture-sensitive conditions [15]. Materials in the nanosize regime bear the highest surface area and the highest number of surface atoms that can be further tuned through the synthesis method employed. This leads to higher catalytic efficiency due to; (i) increasing the contact between the substrate and catalytic center, (ii) offering many catalytic active sites per unit volume as compared to bulk heterogeneous counterpart, (iii) higher zeta potential values, and (iv) due to unique surface chemistry associated with the nanomaterials [18,19].

Consequently, TM NPs based nanocatalysts have emerged as promising alternatives to conventional homogeneous complexes and demonstrated a plethora of applications in medicinal chemistry, electronics, and industrial settings [20–23]. In particular, the application of TM NPs based nanocatalysts in Suzuki, Heck, and Sonogashira cross-coupling reactions has witnessed phenomenal growth in the past decade. For instance, a search on Scopus for articles published since 2010 on the subject matter described in this review is illustrated in Fig. 1.

The data show that in recent years, there has been remarkable growth in the utility of nanocatalysts for coupling reactions. However, issues of their low selectivity, the requirement of harsher reaction conditions, leaching of metal species from the support, and side product formation need to be addressed [24]. Likewise, the controversy of whether it is the surface of NPs (heterogeneous pathway) or the leached metal species in the solution (homogeneous pathway) involved in the catalysis process also requires further attention. As a result, searching for efficient pseudo-homogeneous nanocatalysts with superior performance



Scheme 1. Transition metal NPs catalyzed Suzuki, Heck, and Sonogashira coupling reactions.

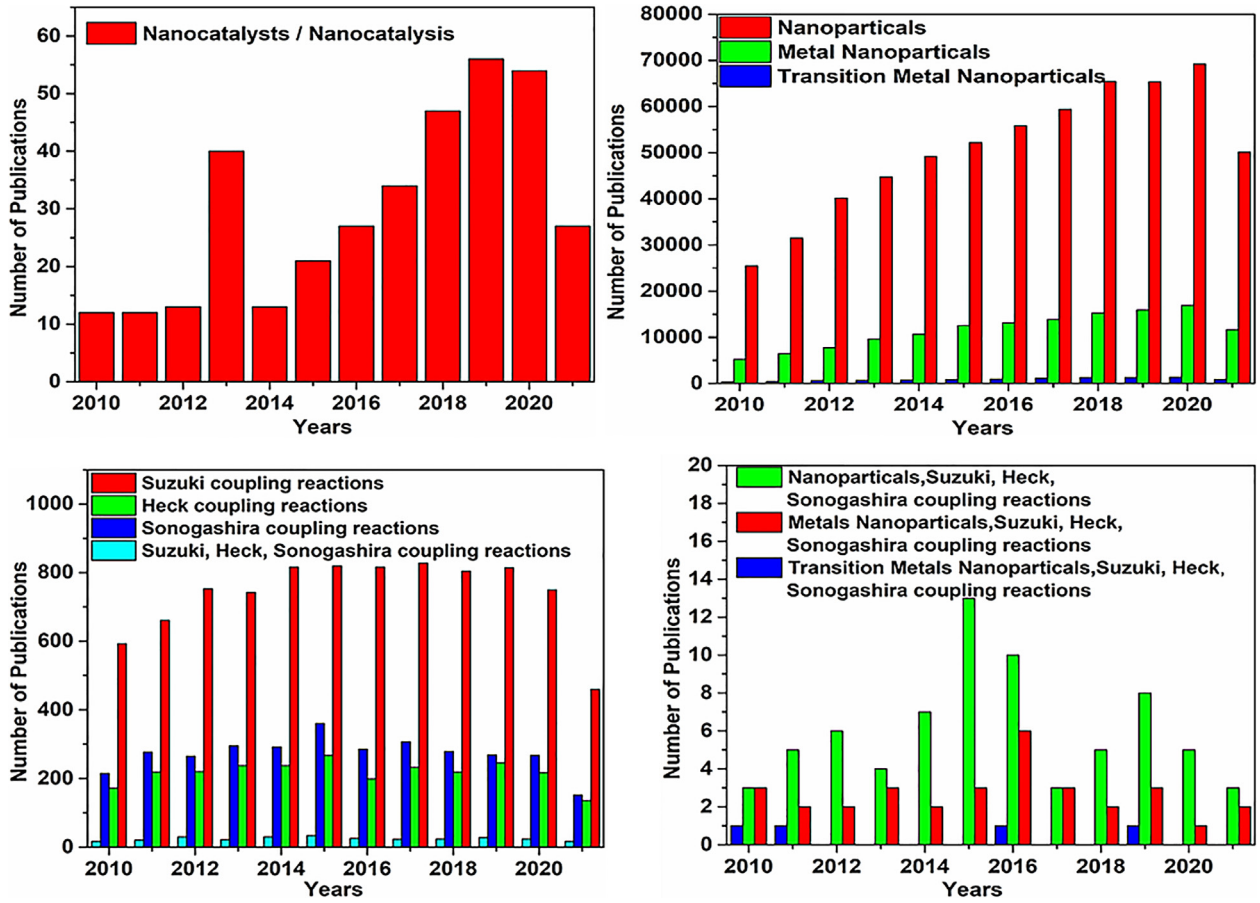


Fig. 1. The number of publications containing different keywords related to our review article since 2010 to date.

to relinquish the liabilities of the homogenous transition metals catalysts remains an ongoing challenge within the field [23,25]. In this contribution, the development progress for synthesizing colloidal nanocatalysts has paved the way to address some of these questions [26–29].

The synthesis of nanocatalysts with controlled size, shapes, and structures has enabled conforming nanocatalysts activities and selectivities. Likewise, the architectures of well-defined nanostructured supporting materials have also made meaningful advancements in the field [30–32]. The use of nanocatalysts of TM NPs is considered a process involving TM NPs on the nanoscale (1 nm to 100 nm in diameter) on solid support (e.g., SiO_2, Al_2O_3, Fe_2O_3, MgO, CeO_2, TiO_2, and others) as solid catalysts and reactants of different phases (e.g., gas or liquid). Generally, the NPs are grown onto a support surface such as alumina, silica, or carbon-based material by impregnation of metal salts, followed by the reduction of these precursors salts to zero-valent metal NPs [33]. These systems are, therefore, up-and-coming for their potential utilization in the industry [34].

Since these supports and resulting nanocatalysts bear a high surface area, even small amounts of dispersed metal NPs onto support can lead to higher efficiency when compared with their bulk counterpart (larger size particles) [35–37]. Moreover, the dexterity to integrate the precise nano-sized (1 to 100 nm (Fig. 2) [38,39] objects into the nanostructured composite affords unprecedented opportunities to generate multiple catalytic interfaces with the elaborated spatial arrangement on a single platform to exploit their synergistic effects [40].

Earlier review articles that summarize applications of nanocatalysts in cross-coupling reactions are relatively specific, such as A^3 (aldehyde, amine, alkyne) coupling reactions [42,43], Pd NPS catalyzed polyols coupling reactions [44], while some are majorly

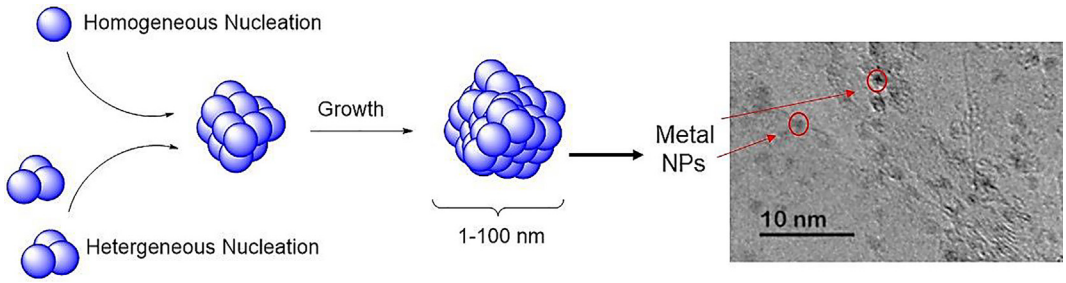


Fig. 2. Growth process of the NPs (red circle represents metal atoms) [41].

concentrated on either composition of the nanocatalyst used or the type of reaction explored. For example, Somorjai et al. reviewed the main concepts of using metal nanoparticles in organic transformation [39]. Some other group-published work includes diverse organic transformations by selective metals (mostly Pd, Pt) nanocatalysts [40–43] with a focus on C—C and/ C–X coupling reactions catalyzed [44,45]. Astruc and co-workers explored the metal NPs catalyzed semi-heterogeneous catalysis as a frontier among heterogeneous and homogeneous catalysis [47]. Likewise, another short review on TM NPs catalyzed C—C bond construction highlights the nanocatalysts role in an aqueous medium for catalytic C—C bond formation [48]. Recently, a detailed review of the polyol synthesis of Pd NPs with different sizes and shapes and their use in cross-coupling and hydrogenation reactions were reported [44]. Recently, Hong et al. reviewed the use of Pd nanoparticles with a special focus on the role of support materials on the in Pd-NPs catalyzed Suzuki, Heck, and Sonogashira cross-coupling reactions [49]. To the best of our knowledge, a comprehensive review dedicated to different TM NPs (Pd, Pt, Cu, Ni, Fe, Au, Ag, and Zn) nanocatalysts is lacking in the literature. Therefore, in this review, we will summarize recent advances in Suzuki, Heck, and Sonogashira cross-coupling reactions catalyzed by TM nanocatalysts, focusing on literature reports majorly spanning from the year 2010 to 2021. We begin with the design of nanocatalysts, with a focus on the size, shape, and role of support material. Then we will give recent advances in transition metals (Pd, Pt, Cu, Ni, Fe, Au, Ag, and Zn) NPs catalyzed C—C bond-forming reactions. Lastly, we will conclude and give the future prospects and main challenges of the field.

2. The design of nanocatalysts

The nanocatalysts catalytic efficiency, selectivity, and recyclability depend on the nanoparticles' size, shape, composition, and assembly, and their instauration with the support. Nanocatalysts may be deigned as nanocrystals, clusters of metals, metals oxide, and nonoxides of various sizes and shapes. These may involve no support materials or may be supported on carbonaceous materials, polymers, metal oxide, or porous materials such as siliceous mesocellular foam [50]. The rational design of nanocatalysts is important to achieve enhanced catalytic activity, chemical and thermal stability, selectivity, and high efficiency [51]. The tailored nanomaterials would necessitate tuning their chemical and physical properties as efficient nanocatalysts through particle size, shape, surface composition, and surface reactivity [52]. It is widely known that electron transfer occurs between the catalyst surface and the reactive species in all key steps of the catalytic reactions, including the reactants adsorption, intermediates diffusion, and the desorption of the products [53]. Thus, tuning the spatial and energy distribution of valence electrons at the catalyst surface would enhance the catalytic efficiency, such as the activation energy barrier, which in turn is linked to the reactivity and selectivity of the catalysts [54,55]. Based on frontier molecular orbital theory, adsorption and bonding of reactants on the surface of the catalyst are dependent upon matching their energy levels to those of the catalyst surfaces as well as the symmetry and spin state of the reactant's molecular or atomic orbitals [56,57]. These properties of nanomaterials can be achieved/tuned by altering their size, lattice framework, morphology, quantum size, and electronic effects from nanopore confinement [58–65]. Likewise, the support's size, shape, and composition are critically important in the design of catalysts [62,63]. In addition, nanocatalysts' thermal and chemical stability is also very important since catalysts can be deactivated because of surface aggregation, leaching, phase changes, and sintering [64,65].

2.1. Size effect

One of the foremost importance of using nanocatalysts is their small size because this is directly linked with other parameters such as high surface area, chemical stability, and reactivity responsible for the enhanced catalytic activities. Therefore, in the past year, enormous work has been dedicated to nanosized catalysts to understand the effect of particle size on their catalytic and electronic properties in numerous transformations, more specifically, coupling reactions [66,67]. When the size of the NPs decreases, the ratio of surface to the bulk atoms increases, denoting that more surface atoms eventually be accessible for the reactions. However, catalytic activity is not often in direct proportion with the increased surface area of smaller NPs [68]. The surface of nanocatalysts with assorted facets can offer diverse binding force strength and configuration, which prolong different active sites, steps, and corners of surface terraces. Therefore, the size and shape of the NPs significantly affect their catalytic performance [69–76]. Likewise, exposure of the crystallographic planes and accessibility of catalyst active sites fitted into the porous structure onto the solid support is also critical for catalytic activity and selectivity [73,74]. The higher performance of nanocatalysts is considered due to the phenomenon of nano-effects, which is believed to result due to structural, quantum size, and electronic effects [58,59,75].

The wet impregnation method is a widely employed technique for preparing transition metal nanocatalysts [76]. However, compared to the solution-phase synthesis of metal nanoparticles, the impregnation method offers poor control over the particle size and size distribution of resulting metal NPs. Nevertheless, the method is still prevalent today due to its scope of scalability and low cost. Other methods for the synthesis of metal nanocatalysts include coprecipitation, which could be in-situ precipitation of the metal and the support using sol–gel protocols [77] or using post-functionalization to deposit the colloidal nanoparticles onto the support [78] or polymer-assisted method that involves polymer as a template to nucleate and stabilize NPs [79,80]. A decrease in the size of the catalyst produces a dominant effect on the surface atoms at edges and corner positions [81,82]. This affects the catalytic activity by creating under-coordinating states on the surfaces. It was found that the fractions of metals on surface atoms having coordination numbers of 4,7,9 in octahedral NPs or 3,6,8 cubic NPs present strong dependences on the size of NPs [83]. The correlation results shown in Fig. 3 illustrate the dependence of catalytic performance on the size NPs.

The general tendency of increased catalytic activity with the particle size decreases indicates that the low coordination number of vertex and edge atoms on the particle surface are active sites for the Suzuki reaction [85] Fig. 4. Further, a series of well-defined homopolymer-stabilized Pd colloids with varying metal particle sizes was utilized to investigate the Heck coupling reaction of olefine and aryl halides [86]. A low reaction rate was obtained when the Pd precursor was utilized without a polymer, implying the reaction doesn't occur through solution-phase catalysis by leached Pd particles. A quantitative relationship was found between the initial reaction rates over various stabilized Pd NPS and the number of low-coordinated (defect) Pd surface atoms calculated using statistics for face-centered cubic (fcc) cuboctahedron [87]. Initial reaction rates did not vary, calculated per number of the defect atoms. The Pd- NPs catalyzed Heck coupling of aryl halides with olefins was thus so-called to be a structure-sensitive reaction with low-coordinated atoms as active sites (Fig. 5). These findings agree with the results for the supported Pd catalyst in the Suzuki coupling reaction [66].

Based on the electronic state viewpoint, the band structure of the metallic state in metal NPs in the range of 1 to 2 nm size behaves as a molecule [88–95], and the molecule exhibits intrinsi-

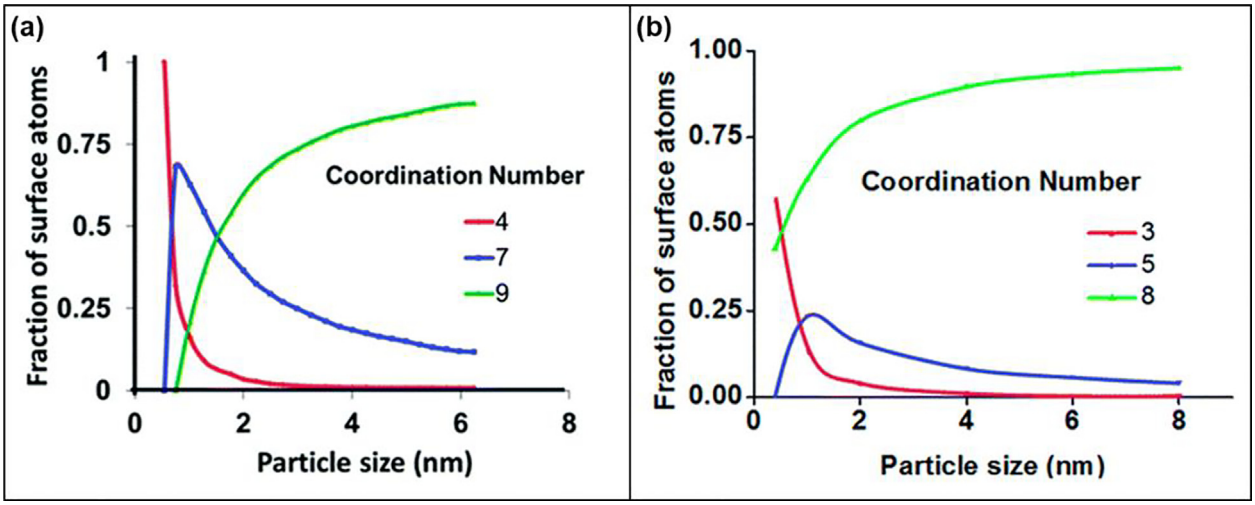


Fig. 3. Diagram of the significance of coordination numbers present on the surface of atoms in NPs assisted catalysis. (a) Plot show fraction having coordination numbers 4, 7, 9 in octahedral nanometer-size particles, (b) Plot shows fraction with coordination numbers 3, 5, 8 in cubic nanometer-size particles [84].

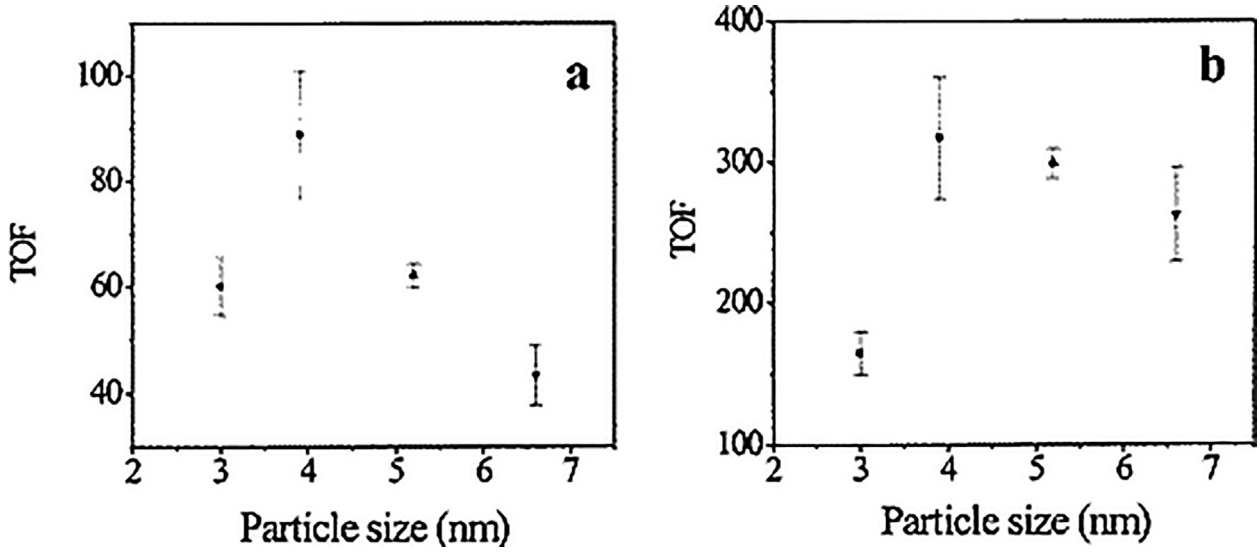


Fig. 4. Turnover frequency as a function of the Pd catalyst particle size for the Suzuki coupling reaction. The TOF is calculated based on (a) total number of surface atoms and (b) total number of vertex and edge surface atoms. Reproduced with permission from ref [85] *Copyright © 2002, American Chemical Society.*

cally better catalytic performance compared to a larger size because it inhabits higher electron density while d band becomes narrowed as closed to the fermi level which suggesting that size is dependent on the electronic environment [73,96–98]. According to the d-band center theory, the catalyst's reactivity depends on the d-band valence electrons near the Fermi level. Consequently, the reactant adsorption, activation, and dissociation are directly related to that metal's d-band center of gravity [95]. Therefore, tuning the spatial and energy distribution of the valence electrons at the surface is important for the reactivity and selectivity of the catalyst. The current literature widely suggests that adsorption energies dependent upon the size of adsorbates on the surface of catalyst atoms rely on the coordination numbers [96–103]. For instance, the surface atoms of catalysts with low orbitals coordination numbers exhibit more adsorption for a specific molecule than higher coordination numbers [84,100]. Computational studies have also suggested that the activation energies of adsorbate for the chemisorption on the surface of Pt NPs catalyst decreases with size. This means activation energies also depend on the size of NPs due to size-dependent cohesive energy, which in turn dictates the size-dependent fraction present on the coordinated surface atoms in nanocatalysts [101].

To conclude, size is one of the key parameters influencing any catalyst's catalytic performance, especially transition metals-based nanocatalysts in different chemical reactions [102]. Its influence is complicated in many cases, such as if the rate-determining step is the dissociation of an intra-molecular bond of a reactant molecule, then it is expected that the metal atom has a lower coordination number and exhibits higher catalytic activity [103–109]. Metal atoms with smaller coordination numbers on the surface have stronger adsorption energy, which could reduce the activation barrier to dissociate the reactant molecules and hence accelerate the catalytic process. The size-dependent catalytic activity is favored when the dissociation step acts as the rate-determining step. While in some cases, this trend is opposite to the above statement, in that case, the metal surface atoms have lower coordination numbers, dramatically dissociate a chemical bond, and tightly bind the dissociated species. Still, unfortunately, this tightly

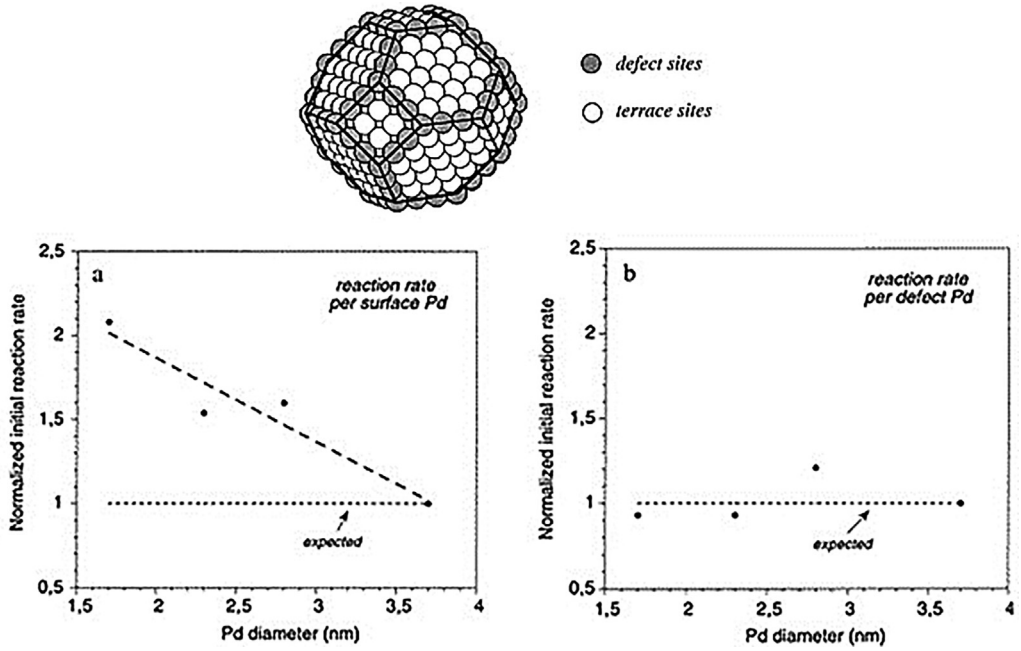


Fig. 5. Structure sensitivity of a heck reaction, initial reaction rates normalized to that of the catalyst with the largest crystallite size and calculated (a) per total surface pd atoms and (b) per defect size atoms. reprinted with permission from ref. [86]. *Copyright © 2002, American Chemical Society.*

binding could poison the surface of the catalyst to slow down the rate of the chemical reaction when coupling with another species during the catalytic reaction cycle [84,103,106,107].

2.2. Shape effect

The shape of NPs plays a significant role in the catalytic performance of nanocatalysts since NPs with different shapes are synthesized by stabilizing certain facets and allowing the other facets to grow [108,109]. This strategy could tune the synthesis with nanocatalysts with exposed active facets and defined atoms on the corners and edges [110,111]. Packing the atoms in nanocatalyst surfaces has been found in various shapes such as spherical, cubic, tetrahedron, octahedron, and truncated octahedron [112–119]. It has been demonstrated that contrasting atomic arrangements and interactions of different crystal faces of the same crystal could have significantly different distributions of valence electrons, resulting in entirely different molecular adsorption and reactivity [116,117]. Thus, the controlled design of the metal NPs crystals with uniform surface facets is necessary for enhanced catalytic activity and selectivity of the product [118]. These surfaces of metal NPs with specific shapes are composed of either pure (100) facets, pure (111), or vicinal surfaces with (730) facets containing denser steps or kinked atoms that result in lowering the coordination number [119]. Therefore, controlled synthesis of these shapes provides the prospects to systematically explore the surface structure to tune the catalytic activity and selectivity of the catalytic reactions in organic synthesis [84,111]. The low-index facets such as (100), (110), and (111) of magnetic nanoparticles (MNPs) are preferably exposed due to the lower surface energy and hence more stability. The variations in structural aspects stem mainly from the differences in the atomic packing on the catalyst surfaces [120]. Therefore, the structural variations between different crystallographic surfaces intrinsically influence the dissociation, adsorption, coupling, and interactions of the adsorbed intermediates in the catalytic reactions [84].

A good fraction of the catalysis research was devoted to tailoring metal deposits' shape on oxide surfaces. One parameter that controls the particle geometry is the metal-support interaction, which is typically small for wide-gap oxides, causing the metals to grow in a 3D fashion [121]. Various approaches were developed to modify this parameter, for instance, by introducing defects and species (OH functionalities) into the oxide surface or using polar oxide terminations with high metal oxide adhesion[122]. Alternatively, electron-transfer processes across the metal oxide interface may be exploited to alter the growth morphology of metals. This approach has been demonstrated for thin fil oxides on metal support first [123]. The shapes of metal nanoparticles, particularly in wet chemistry synthesis, can be controlled in the twilight zone of kinetic and thermodynamic control [115]. This is mostly controlled by the capping ligands, solvents, and temperature by controlling the nucleation and growth steps [124–130]. El-Sayed et al. reported that Pt NPs have cubic, tetrahedral, and spherical spiracles shapes (Fig. 6). These shapes were controlled by altering the concentration of capping materials to the concentration of Pt cations used in the reductive hydrogen method at room temperature [127]. Later on, the same research group reported the Pt NPs with various shapes as nanocatalysts for electron transfer reactions [66,128]. The tetrahedral-shaped Pt NPs showed excellent catalytic activity with the lowest activation energy (14 kJ mol^{-1}), whereas cubic-shaped Pt NPs exhibited low catalytic activity, having the highest activation energy (26.4 kJ mol^{-1}). The sphericle-shaped Pt NPs displayed moderate catalytic activity due to exposed facets and fewer catalytically active atoms on the surface of NPs. Thus, tetrahedral-shaped Pt NPs exhibiting (111) facets had a large fraction of atoms on the edges and corners of the surface. On the other hand, cubic-shaped Pt NPs contained (100) facets on the surface with smaller fractions of atoms on the edges and corners of the surfaces. Due to these exposed facets and catalytically active atoms on the surface of tetrahedral NPs, the tetrahedral Pt NPs proved more active than cubic Pt NPs [128].

The shape dependence of NPs on the reactivity in C—C bond formation, e.g., in the Suzuki reaction, has been demonstrated by Bokhoven et al. [74]. The catalytic activity tends to increase from spherical to tetrahedral shapes of Pt NPs, which was linked to the (111) facets of tetrahedral shapes that bear the maximum

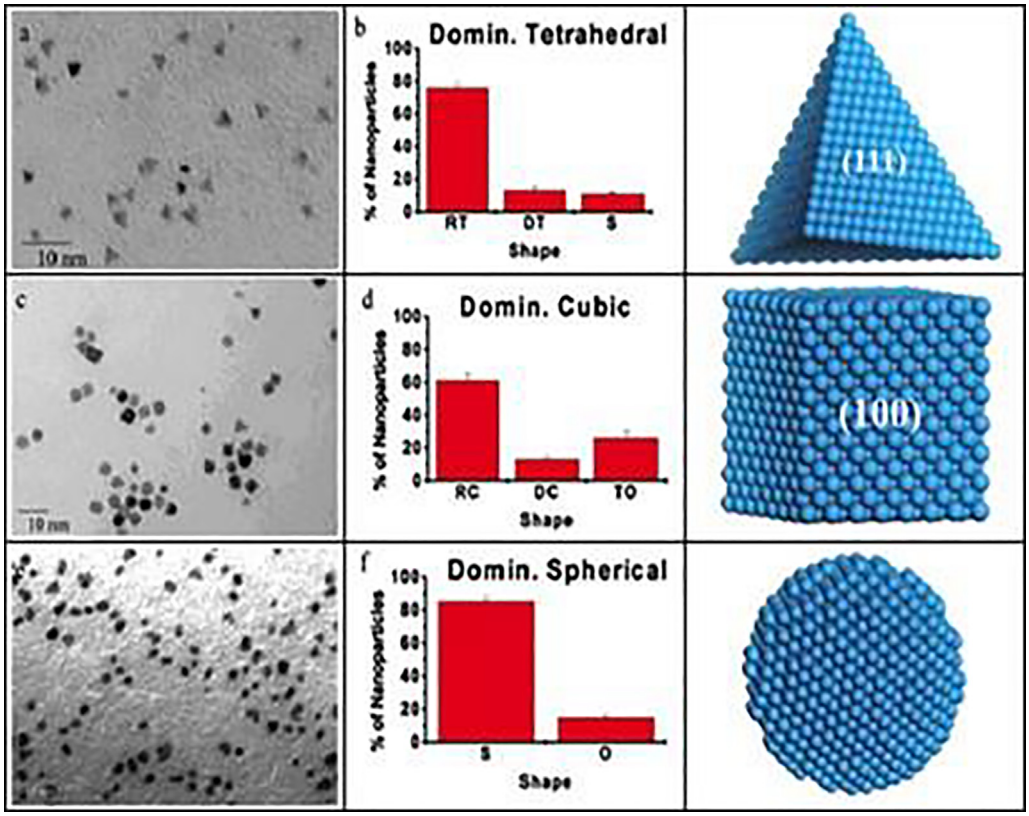


Fig. 6. TEM images of tetrahedral, cubic, and spherical shapes Pt NPs. Figure adapted from the American Chemical Society [128].

catalytically active sites on the surface of NPs. Likewise, Huang reported the synthesis of Pd nanorods and branched Pd nanostructures for the C—C coupling reaction [129,130]. The study concluded that both nanorods and branched NPs were proved to be very active catalysts with comparable chemical stability and recyclability. However, the nanorods had a little edge (92 % yield) over star shape (90 % yield).

2.3. Support materials

In the case of supported metal catalysis, the support materials have been considered to be inert, and their prime role was to increase the stability of the metal particles by immobilizing and dispersing them to improve the active surface area. Later, the Tauster group introduced the term strong metal-support interaction to elaborate on the unexpected chemisorption properties of H_2 on noble metal NPs supported by TiO_2 [63]. Further, the importance of the support materials has been explored at a fundamental level, commonly utilizing model catalysts. In short, the role of the support materials in catalysis is somewhat more complex and yet to be explored. Support material can play direct or indirect roles during a catalytic transformation. These provide precise defect sites onto which the metal NPs can be immobilized in the case of metastable particles [131]. In addition, the support can enable the electronic interactions between metal NPs and provide functionalities to the nanomaterials, such as acidic sites or basic sites that improve the stability, activity, and selectivity of the catalyst [132,133].

Earlier studies have shown that the catalysts inherent activity and selectivity may vary strongly, contingent on the size of the metal NPs (>7 nm in diameter). Yet the size is not solely responsible for varying catalyst efficiencies. The choice of catalyst support is captious even for materials having the same geometry, for which mass transport differences can be ruled out [13]. This frequently turns out that some support materials catalyze unwanted side transformations, in many cases, the clarification is less frivolous [134]. In some cases, the NPs catalytic activity decreased upon recyclability. Therefore, a stabilizer such as a polymer ligand and surfactant was used to overcome the mentioned issues [135]. A series of supporting materials such as CNTs [136], colloidal support [137], silica [137], metal NPs support [138], polymers [139], carbon [140], and graphene [141] have been reported. On the other hand, using magnetic NPs as catalyst support could also enable the effective separation of catalysts from the reaction mixture with the help of an external magnet [142]. In summary, the nanostructured support materials hosting the active metal NPs nanocatalysts significantly reduce the overall cost, provide simple and scalable preparation, and easy filtration for efficient recycling [143].

3. Transition metals NPs catalyzed CC bond-forming reactions

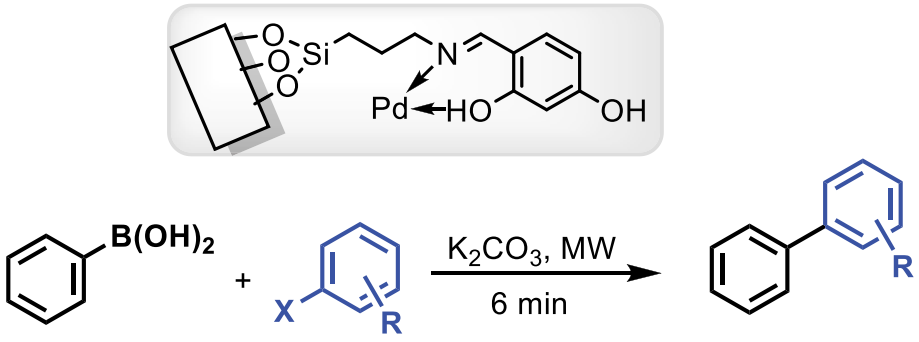
3.1. Pd NPs catalyzed C—C bond-forming reactions

Palladium nanoparticles are utilized in various C—C coupling reactions [144,145], which are the essential methodologies of modern organic synthesis. A one-step approach can achieve various aryl derivatives with high yield and selectivity without isolating the intermediate products. Pd NPs catalyzed cross-coupling reactions are well documented by different research groups [46,146,147]. The Pd NPs can be immobilized on different supports. In this context, Liew et al. recently reported the immobilization of Pd NPs onto fluorine-doped tin oxide (FTO) support. The nanocatalyst was employed in Suzuki coupling reactions of aryl halides and phenylboronic acid to construct biphenyl in reasonably high yields (up to 80 %). The Pd nanocatalysts showed a high degree of chemical stability [148]. In another study, Mahmoud et al.

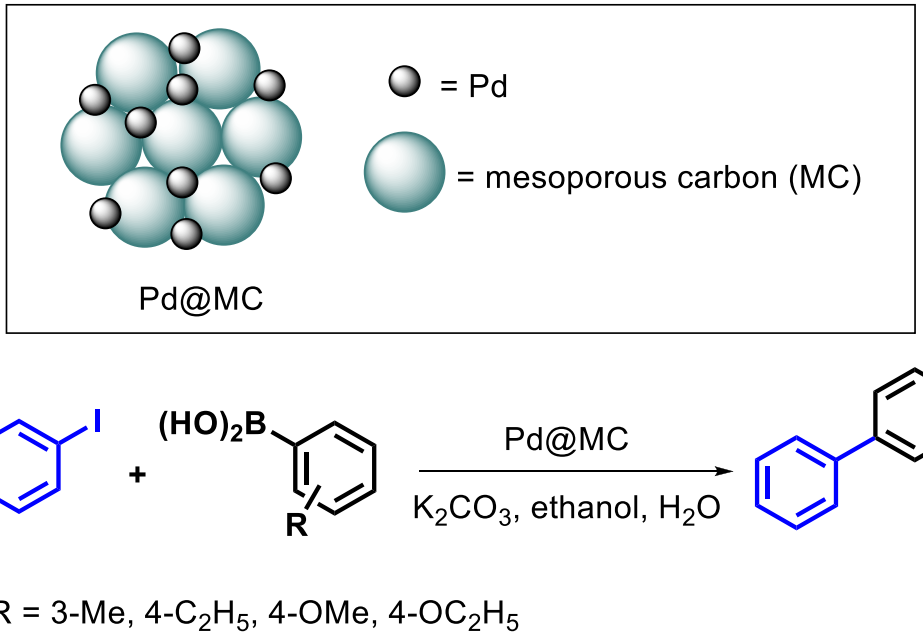
immobilized the Pd NPs on Schiff base-modified boehmite particles (Pd NPs@Sch-boehmite) using ethanol as a reducing agent. The as-prepared nanocatalyst was utilized as a retrievable catalyst for Suzuki coupling reaction under microwave irradiation in solvent-free conditions to produce biphenyls in excellent yields (up to 96 %). The nanocatalysts proved efficient as the reactions required a very low catalyst loading (0.005 mol%). Moreover, the Boehmite surface was proved to be an excellent stabilizer for supporting Pd NPs, since nanocatalysts could be reused after repeated 8 cycles without a considerable loss of catalytic activity (Scheme 2) [149].

To further provide stability to the catalyst, the support materials should have more anchoring sites, such as nitrogen-containing functional groups (e.g., amino groups) or nitrogen-doped carbon-based materials to immobilize metal NPs. This covalent binding of metal NPs increases the stability of the prepared catalyst. In this contribution, Liu et al. encapsulated the Pd NPs in a nanoscale amine-functionalized tubular silica support to form a physically confined nanocatalyst with a high specific surface area. These structural features were favourable for good dispersion of Pd NPs, simple and fast mass transport of reactants, and easy access of reactants to active sites. Finally, the nanocatalyst was utilized for the Suzuki coupling reaction with 98 % yield of the coupling product (Scheme 3) [150].

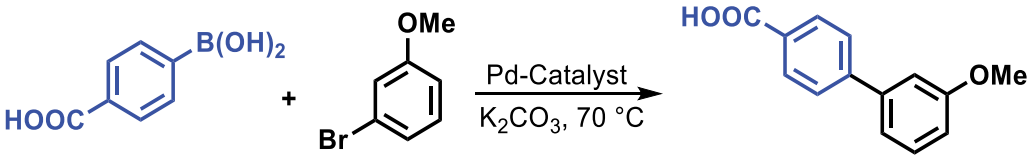
Nanostructure alumina was treated with different quantities of $PdCl_2$(1,5-cyclooctadiene) to promote the formation of supported Pd NPs, which were then reduced in situ in a single step to give dispersed Pd NPs as catalysts [151]. The prepared materials were tested in a specific C—C Suzuki coupling reaction (Scheme 4). Upon recyclability, a much higher loss was observed in the case of carboxyphenylboronic acid (ca. 55 %) compared to 4-vinylphenylboronic acid (ca. 10 %). This is maybe due to two different effects, first the possible agglomeration of the NPs after the first cycle, which has been previously reported [152]. In contrast, the second reason may be an additional effect related to the carboxy functional group. A probable explanation is that the formation of stable intermediate species such as (Ar-Pd-OOC-Ar) (Pd (II)



Scheme 2. Schematic presentation of Pd NPs@Schiffboehmite application in Suzuki coupling reaction.



Scheme 3. Pd NPs@MC catalyzed Suzuki coupling reaction.



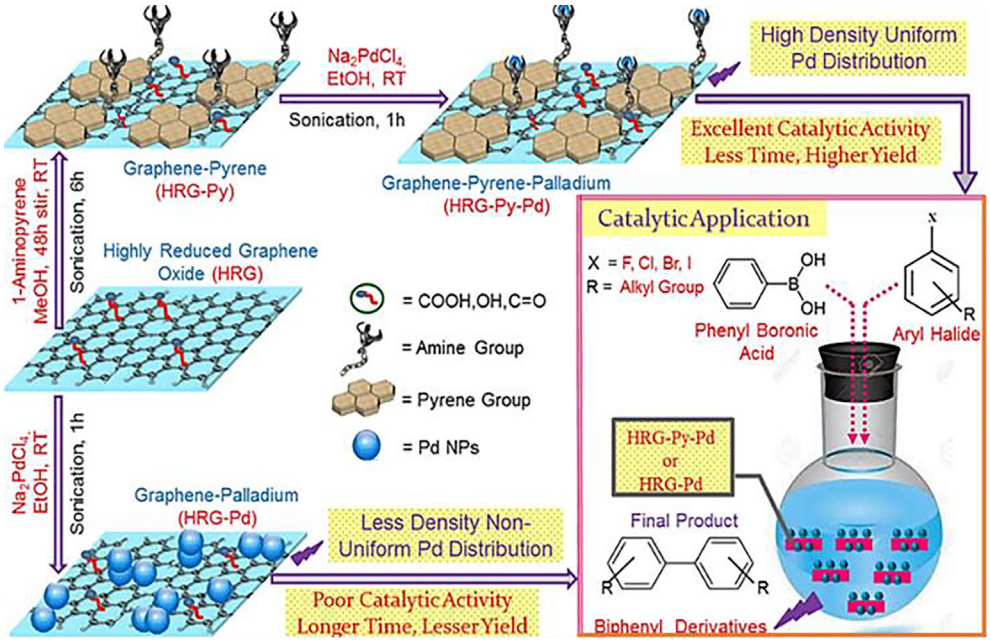
Scheme 4. Pd NPs on Alumina for the Suzuki coupling reaction.

species), which are generated during the catalytic transformation and which may partially deactivate the Pd NPs leading to lower conversions in the second and following catalytic cycles as the active species should be based on Pd (0). It is important to note that the formation of these species is impossible in the case of 4-vinylphenylboronic acid as this compound does not have carboxy groups.

Recently, our group has reported Pd NPs functionalized onto highly reduced graphene oxide (HRG) as nanocatalyst for the Suzuki coupling reactions in water (Scheme 5) [153]. The nanocatalysts system did not require any pre-activation or inert reaction conditions. 1-aminopyrene stabilized the Pd NPs on the surface of graphene. The density of the Pd NPs was increased through aminopyrene functionalization, where the amine group provides the nucleation site for Pd NPs. Likewise, the amino groups also helped to avoid aggregation and agglomeration of the NPs. The nanocatalyst was used for the coupling reaction of various aryl halides with pyreneboronic acid to yield 1-phenylpyrene in excellent conversion (up to 100 %).

The turnover frequency (TOF) values for HRG-Pd and HRG-Py-Pd were calculated and compared, which were 197 h^{-1} and 1000 h^{-1}, respectively. The TOF for HRG-Py-Pd is 5 folds greater than the HRG-Pd catalyst. However, it is found that the turnover number (TON) values for HRG-Py-Pd were not too much higher than HRG-Pd [153]. To investigate the substituent effect, the performance of as-prepared nanocatalyst HRG-Py-Pd was evaluated for the Suzuki coupling reaction of bromo, chloro, and iodo benzene and other several types of substituted aryl halides having both the electron donating groups (EDG) and electron-withdrawing groups (EWG) with variation of phenyl boronic acid in aqueous conditions. In the first step, it was investigated that (Fig. 7) the iodo substituted benzene needed a minimum 5 min time for complete conversion of reactants into products using HRG- Pd as catalyst, while for the HRG-Py-Pd nanocatalyst the reaction was taken up to 20 min. The same types of patterns were observed for chloro and bromo-substituted benzene. Besides these, the catalytic efficiency of HRG-Py-Pd nanocatalyst was tested for other various substituted aryl halides such as 4′-bromoacetophenone, 4-chlorobenzenephenone and 4-bromoanisol. The substituted aryl halides have slightly shown a lesser amount of conversion efficiency than the non-substituted aryl halides due to steric hindrance present in these substituents. Furthermore, EDG yielded a higher conversion among aryl halide compounds than aryl halides substituents. For example, 4-bromoanisole containing EDG (methoxy group) demonstrated 100 % conversion efficiency compared with 4-chlorobenzophenone consisting of (78.9 % conversion) as shown below in Fig. 7 [153].

Erami et al. utilized commercial COOH-modified graphene to support the Pd NPs. The NPs were presumably formed by the reduction of $PdCl_2$(1,5-cyclopctadiene) [154]. Pd-doped graphene has shown interesting catalytic behavior in the Suzuki coupling reaction of fluorinated aryls with TOF values up to 67 h^{-1}. Furthermore, the recyclability of the catalyst at a short three hours reaction time shows a high loss of activity in the catalytic tests from 90 % to 47 % after the 5th cycle. Nevertheless, this material exhibited a reasonable degree of recyclability with a low deactivation of less than 8 % after up to five catalytic cycles after 48 h, possibly due to the longer reaction time compensating for the loss of activity. Nishina et al. reported the synthesis of Pd NPs supported onto



Scheme 5. Preparation of Pd NPs doped highly reduced graphene composites and their catalytic applications.

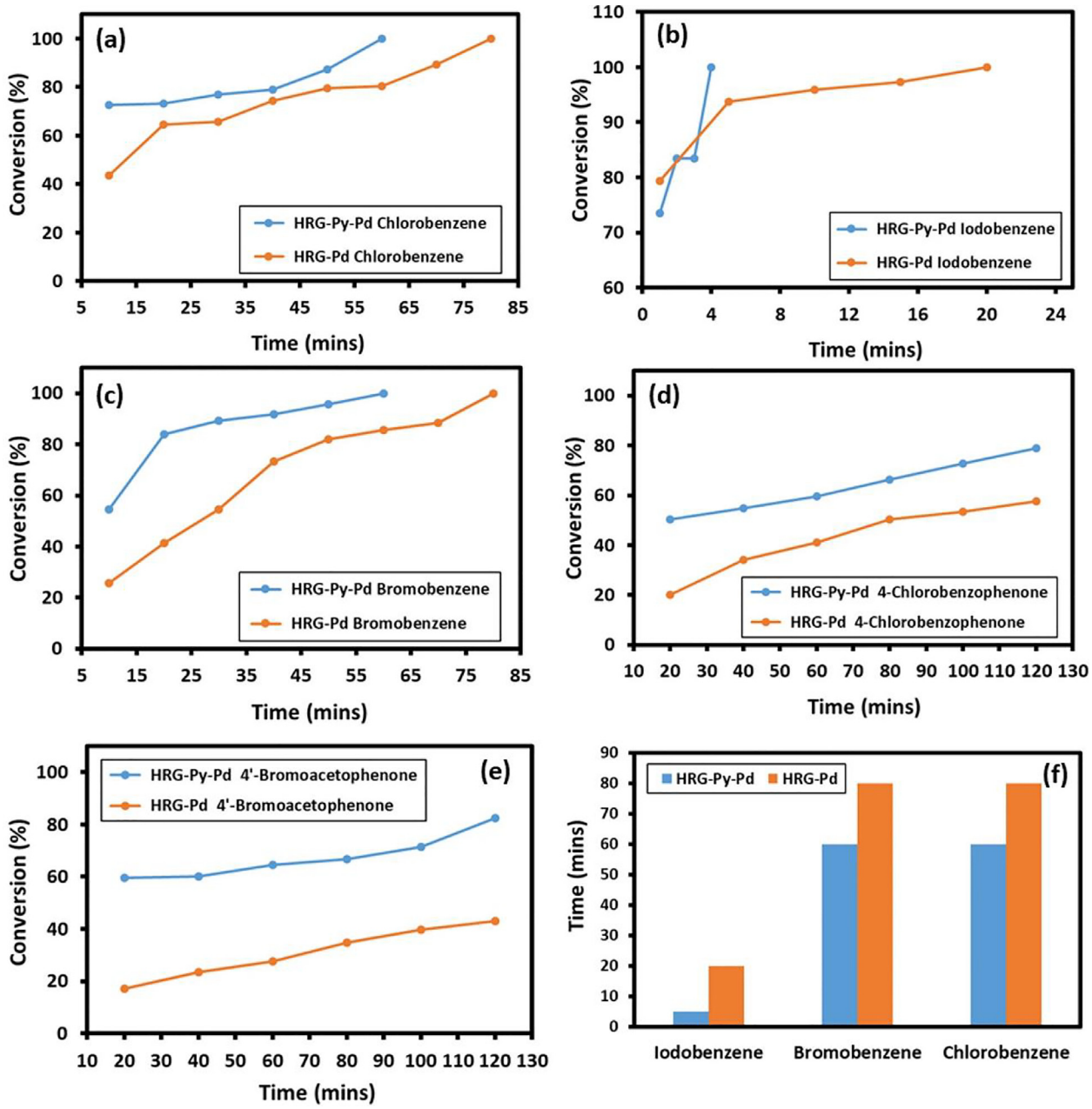
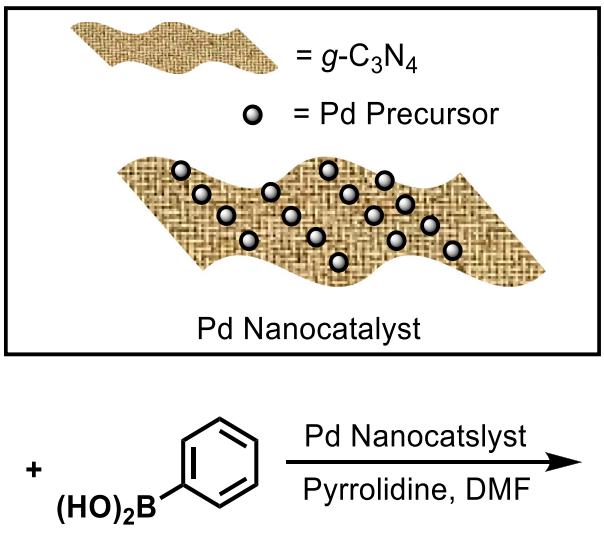


Fig. 7. Time dependent conversion efficiency of the Suzuki reaction employing HRG-Py-Pd and HRG-Pd for various substrates using GC analysis: (a) chlorobenzene, (b) iodobenzene and (c) bromobenzene, (d) 4-chlorobenzophenone, (e) 4′-bromoacetophenone (f) comparison of the conversion (100%) of product using HRG-Py-Pd and HRG-Pd with respect to time [153].

single-layer graphene oxide via reducing media in the presence of ethanol and water as solvents to enhance the TOF value. The as-prepared catalyst was employed as a heterogeneous catalyst in synthesizing biphenyls in high yields [155]. The nanocatalysts demonstrated high robustness since only 0.10 ppm of leaching of Pd was detected. Moreover, the nanocatalyst demonstrated enhanced catalytic activity with a turnover number (TON) and TOF (h^{-1}) of 237,000 [156]. The leaching was further controlled by 0.25 wt% Pd coordination with the support such as halloysite functionalized with 2-(diphenyl-phosphino)ethyltriethoxysilane [157]. In another study, supported Pd NPs on SBA-15 have been reported (with a maximum loading of Pd of 12.80 wt% on SBA-15). The described method does not need the use of functionalized mesoporous silica. The SBA-15-Pd catalyst has been studied in Suzuki coupling reactions of phenyl boronic acids with aryl and heteroaryl halides. After the 5th recycling study, the hybrid material exhibited higher conversion rates and good recyclability, and Pd leaching was not detected. Also, hybrid materials SBA-15-Pd have been carried out in vitro against human cancer cells and demonstrated very high cytotoxic activity [152]. Sasidharan et al. reported direct immobilization of Pd NPs onto the surface of polymeric mesoporous nitrogen-rich C_3N_4 support. The as-synthesized nanocatalyst was used for the Suzuki cross-coupling reaction. The Pd NPs were locked in a mesoporous 3D π network of C_3N_4 without stabilizing agents (Scheme 6) [158].

Kong group reported the easiest way to synthesize the Pd NPs over nitrogen-rich triazine-assisted porous organic polyamines (PdNPs@POPa). The narrow size NPs distribution over porous organic polyamines in the as-prepared nanocatalyst shows a remarkable reductive catalytic activity of nitrobenzene and its

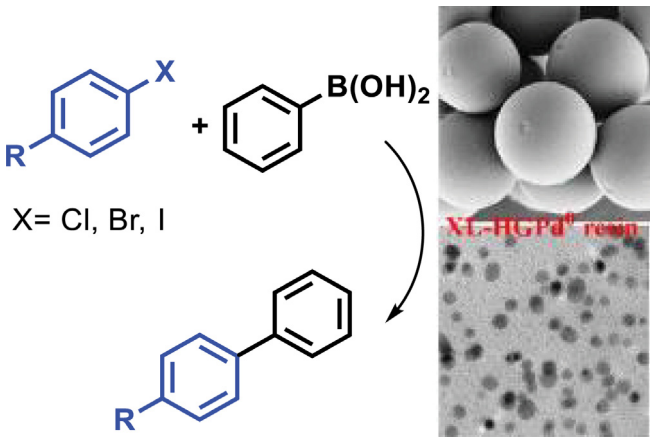


Scheme 6. Illustrating of Pd(0)-C_3N_4 heterogeneous catalyst preparation utilized in Suzuki reaction.

derivatives during the Suzuki coupling reaction. Profiting from POPa the as-used Pd-based nanocatalyst PdNPs@POPa can be easily recovered and recycled with negligible loss of catalytic activity [159]. Liew group synthesized highly dispersed Pd NPs immobilized onto HypoGel resins using an extensive cross-linking method. The further reduction of Pd precursor resulted in nanoparticles with an average size of about 8 nm. In addition, these nanoparticles did not require any additional ligands. The as-prepared heterogeneous catalyst exhibited excellent catalytic activity in the preparation of diverse biaryls from a series of aryl halides and phenylboronic acid. The catalyst showed enhanced mechanical and chemical stability with only 0.25 % weight loss of Pd (0.55 ± 0. 02 ppm) from the support after the first cycle reaction. Moreover, the recovered catalyst demonstrated excellent reusability up to five cycles (Scheme 7) [160].

Veisi et al. reported the synthesis of magnetically divisible Pd NPs supported on coated silica on Fe_2O_3 NPs as a catalyst for the Suzuki coupling reaction (Scheme 8). Fe_2O_3 NPs coated with silica loaded with Ag and Pd catalysts showed good catalytic activities compared with commercially available Pd/C catalysts [161,162].

Another class of nanostructured materials is metal–organic frameworks (MOF) which have many extraordinary properties such as a well-defined pore structure, high surface area, and stable physicochemical features and, most importantly, can be modified

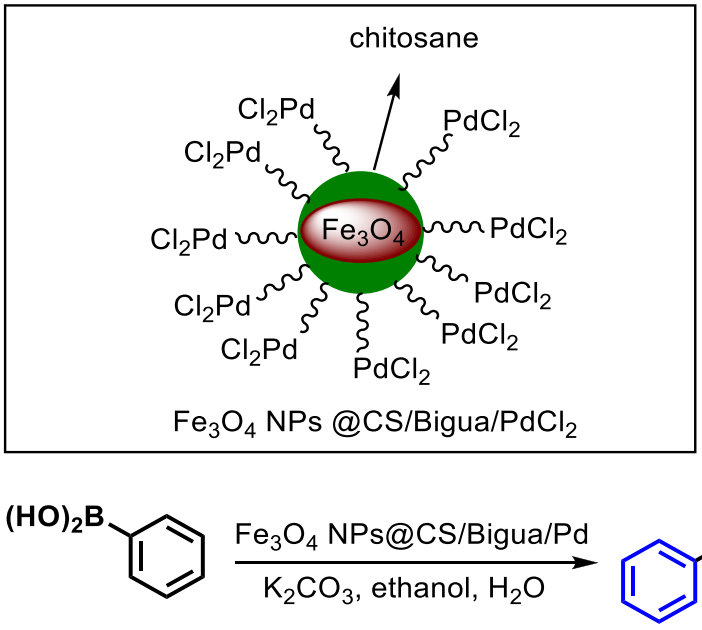


Scheme 7. Catalytic scheme of PdNPs (XL-HGPd) by a cross-linking method [160].

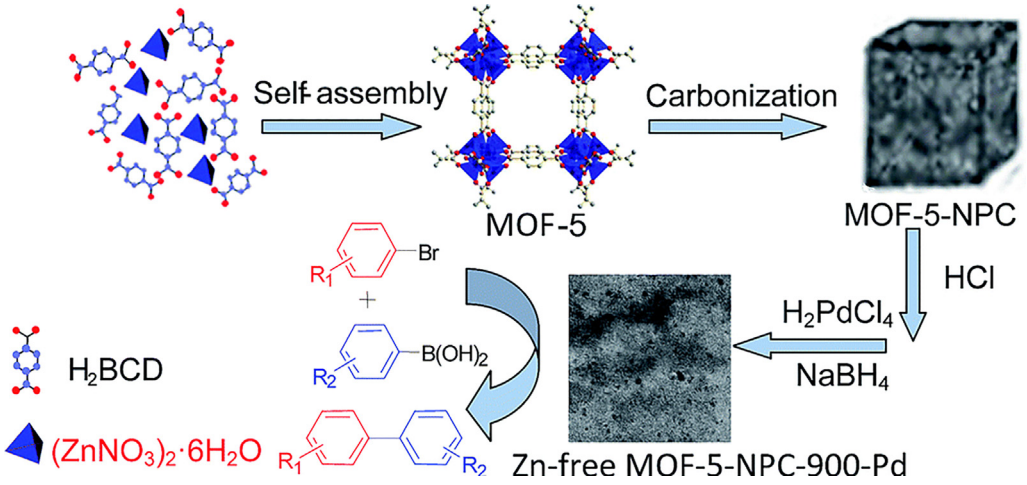
with various functional groups through the stabilization of various metal/metal oxide NPS with controllable sizes [163,164].

Zhang et al. described a highly porous MOF-5 with well-defined cavities/channels loaded with $[Pd(C_3H_5)(C_5H_5)]$ as a metal – organic precursor and assembled by the metal – organic chemical vapor deposition method [165]. The precursor adsorption process was used to achieve the Pd NPs ingraining into the cavities of MOF-5, followed by hydrogenolysis of the inclusion compound because of the loss of the C_3H_5 ligand and interaction of Pd and benzene-1,4-dicarboxylate linker in MOF-5. The small quantity of the prepared heterogeneous catalyst (0.17 wt%) revealed an excellent performance in the Suzuki coupling reaction using K_2CO_3 in the solvent of DMF at 90 °C for 20 h. The catalyst, $[Pd(C_3H_5)(C_5H_5)]$@MOF-5 nanocomposite, could be separated from the reaction media but exhibited an inappropriate catalytic activity in the subsequent runs due to the instability of the pore structure of the MOF support. Further, to get a more stable and active MOF-based catalyst, the Wang group utilized one-step direct carbonization of a MOF to get zinc-free MOF-5 without any additional carbon precursors [166]. Followed by the Pd NPS immobilization on the MOF-5 derived nanoporous carbon by an impregnation method coupled with subsequent reduction with $NaBH_4$. The catalyst was then utilized for the Suzuki coupling reactions and showed high catalytic efficiency with a high yield ranging from 90 % to 99 % under mild reaction conditions (Scheme 9). The catalyst was recovered and recycled using the reaction of bromobenzene and benzene boronic acid as the representative reactants in the presence of 0.1 mol% of Zn-free MOF-5-Pd nanoporous carbon. Upon recyclability, the Pd amount was decreased from 4.85 % to 4.4 %, which may lead to the disability of the catalyst.

Zou et al. reported that Pd NPs supported MIL-101-NH_2(Cr) as an efficient heterogeneous nanocatalyst for the Suzuki coupling reactions at room temperature [167]. Further, the author compared cations such as cesium and potassium and anions like carbonate and fluoride of the utilized bases for their effect on the catalytic activity of the catalyst, catalyst crystallinity, and Pd NPs distributions. Similar to the previous work, the as-prepared Pd@MIL-101-NH_2(Cr) catalyst was degraded with the carbonates, while it could keep its porosity and crystallinity with fluorides. Indeed, numerous MOFs-supported Pd NPs enhanced the Suzuki coupling methods under mild conditions [167]. In 2013, the Pascanu group utilized the Pd NPs-supported MIL-101Cr – NH_2 for the Suzuki coupling reaction with excellent activity and recyclability [168]. The size

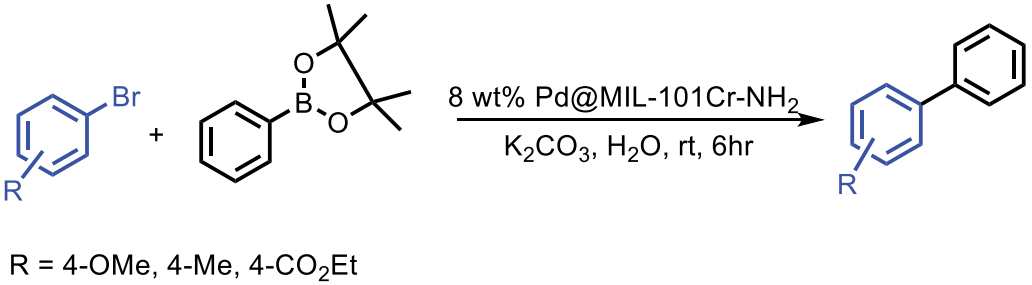


Scheme 8. The scheme showed Pd-supported silica-coated Fe_2O_3 NPs as a catalyst.



Scheme 9. The catalyst preparation process and the Suzuki coupling reaction. Reproduced from Ref. [166]. with permission from the Royal Society of Chemistry.

effect, distribution, and also Pd loading (4, 8, 12, and 16 wt%) were examined on the catalytic activity for the prepared Pd@MIL-101Cr – NH_2 nanocomposite catalyst, and the best catalytic activity and reusability were achieved for 8 wt% Pd NPs of 2.6 nm average size. The obtained Pd@MIL-101Cr – NH_2 was used successfully in the Suzuki coupling reaction of (hetero)aryl bromides with pinacol phenylboronate in the presence of K_2CO_3 in H_2O or H_2O/EtOH at room temperature (Scheme 10), with the reactions being completed in an aerial atmosphere for 6 h. The catalyst was separated from the reaction mixture and then recycled for 10th cycles without any alteration in activity. Notably, the solvent and base's nature greatly affected the Suzuki reaction's efficiency in this system.



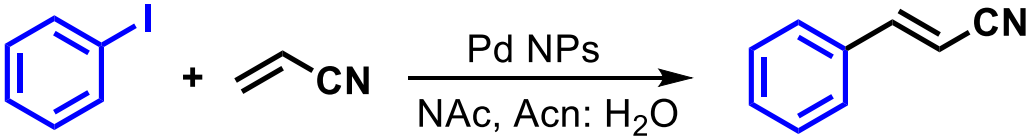
Scheme 10. Suzuki coupling reactions catalyzed by Pd@MIL-101CR-NH_2.

The size and shape of the Pd NPs can be controlled by changing the reaction condition of the corresponding solution parameters such as stabilizing ligand, pH, temperature, and time [169]. These parameters also tune the morphologies of the resulting Pd nanostructure, e.g., spherical, dendritic, rectangular, and triangular-shaped Pd [170]. Depending upon morphology, these nanostructured nanocatalysts can alter the activity and selectivity of the reaction product. For example, Kamal et al. synthesized the Pd NPs supported onto carbon spheres. The nanocatalyst was utilized for the Heck coupling reaction, resulting in significantly high activity (98 % yield of the product) (Scheme 11) [171]. The dispersion of the Pd NPs on the carbon surface leads to high activity and selectivity [172].

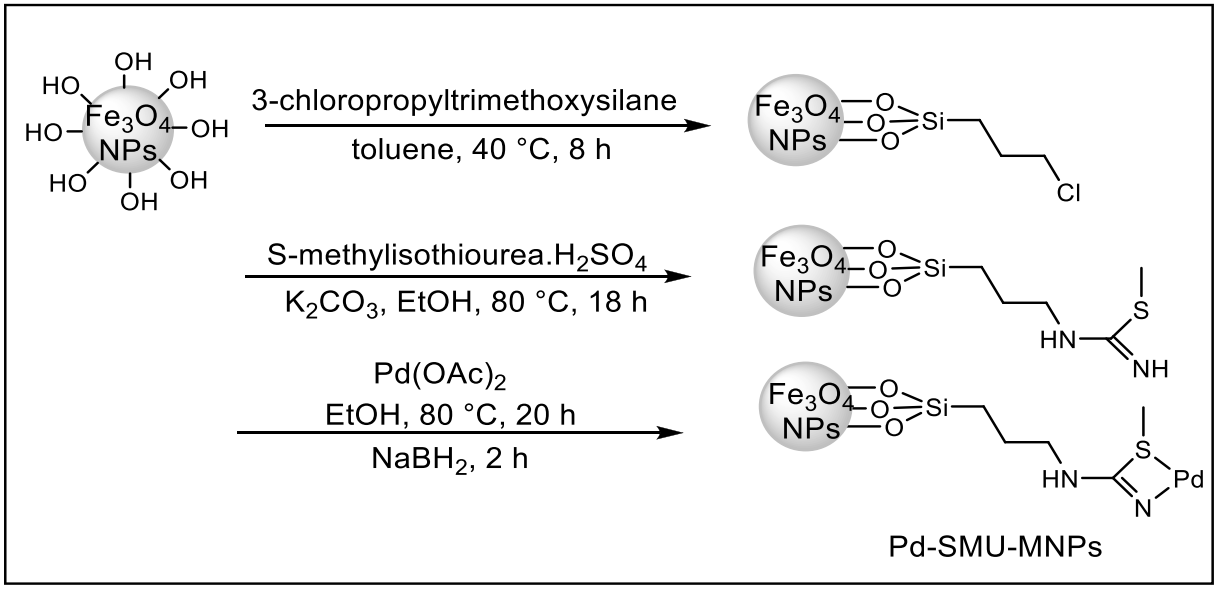
In quest of simplifying nanocatalyst recovery and reusability, Arash et al. prepared the heterogeneous Pd nanoparticles bound to Fe_3O_4 NPs through -*S*-methylisothiourea ligand (Pd-SMU-MNPs) [177]. The nanocatalyst was then employed for the Suzuki cross-coupling reaction between diverse aryl halides and phenylboronic acid to produce a variety of biaryls in reasonably good yields (up to 97 %). Using magnetic NPs (MNPs) as support was attractive since the nanocatalysts could be rapidly recovered from the reaction contents using an external magnet. The nanocatalyst exhibited considerable catalytic activity (95 % yield of the product) without a significant loss for consecutive 8 runs (Scheme 12) [173].

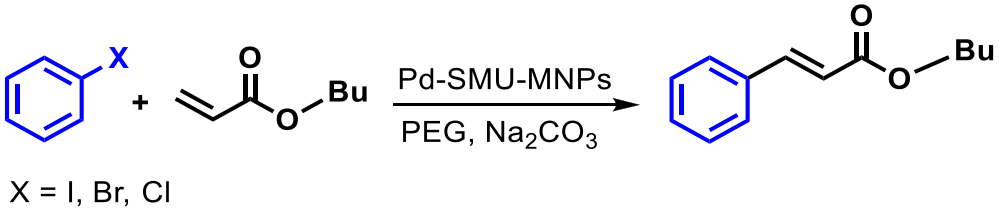
Naghipour et al. also adopted a similar strategy by coating magnetic nanoparticles (MNPs) core with a shell of chitosan, which was then tailored with Schiff base ligand followed by immobilization of Pd NPs to produce heterogeneous nanocatalyst (Fe_3O_4@CS-Schiff base Pd). The as-prepared nanocatalyst was used in Suzuki and Heck's cross-coupling reactions to produce diverse biphenyls as well as diphenylethene derivatives in yields of 98 % (Scheme 13) [174]. As a green solvent, an enhanced catalytic activity and product conversion for the Suzuki coupling reaction was achieved in polyethylene glycol (PEG). In the case of the Heck coupling reaction, DMF was found to be the better solvent for a higher product yield of 98 %. The catalytic activity of the recovered catalyst from the reaction was almost unchanged after five cycles.

Pitchumani et al. incorporated cube shape Pd NPs (Pd cNPs) with [96] surface facets exposed on the surface. These were immobilized onto carbon@Fe_3O_4 nanospheres (C@Fe_3O_4) to prepare heterogeneous catalysts (Pd cNPs/C@Fe_3O_4) for Heck coupling reactions (Fig. 8) [175]. The nanocatalysts were highly efficient, providing the corresponding diphenylethene products in excellent yields (99 %). The development of carbon shells on Fe_3O_4 was important to preserve the Fe_3O_4 NPs or prevent them from oxidizing by acids and oxygen. The C@Fe_3O_4 turned out to be excellent support for the anchoring of catalytically active metal species, resulting in structurally stable catalytic sites. The high catalytic

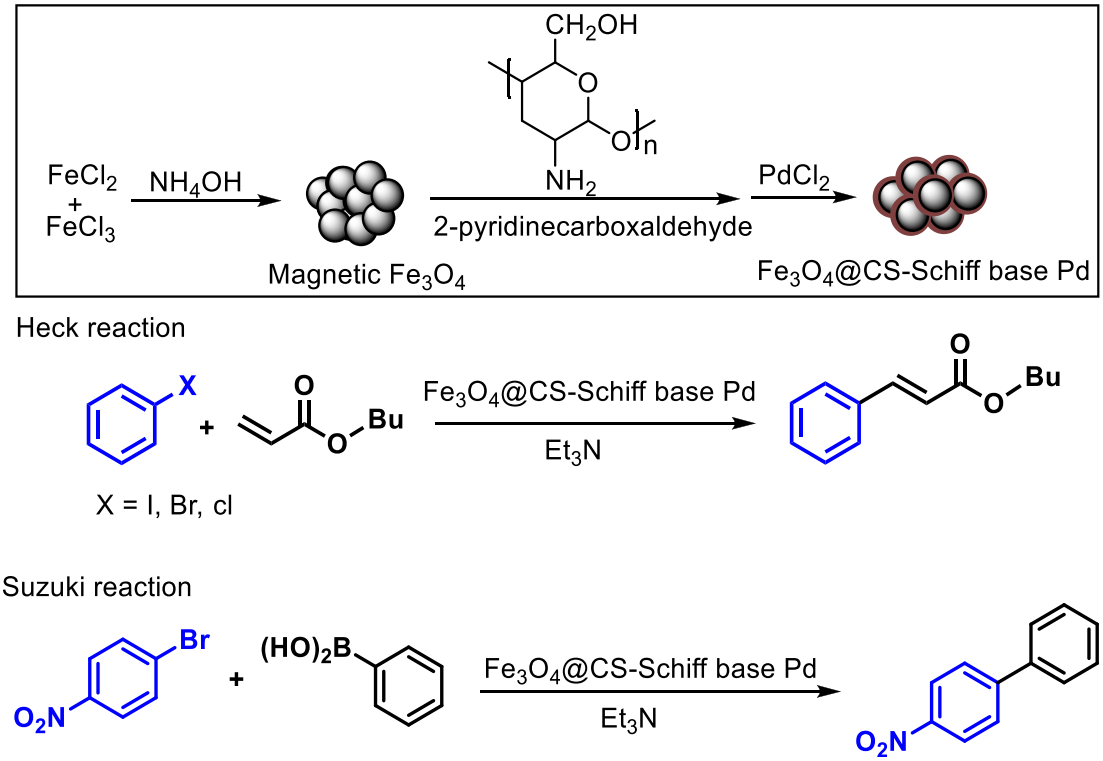


Scheme 11. Heck reaction catalyzed by Pd NPs.





Scheme 12. Pd-SMU-MNP catalyzed Heck reaction.



Scheme 13. Fe_3O_4@CS-Schiff base Pd catalyzed Heck and Suzuki coupling reaction.

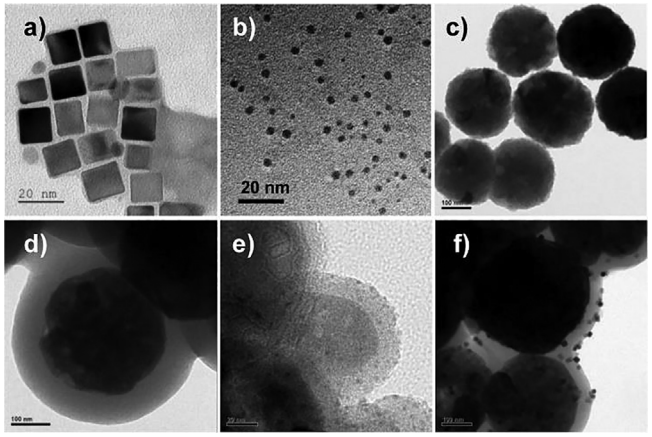
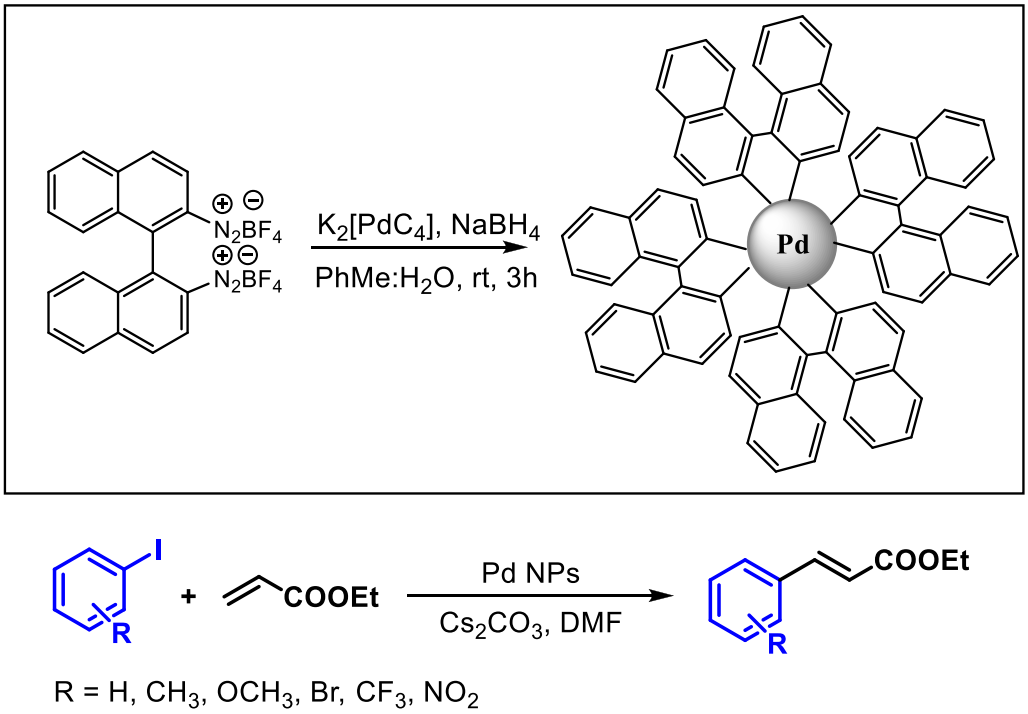


Fig. 8. TEM images of (a) Pd cubical NPs, (b) Pd spherical NPs, (c) Fe_3O_4, (d) C@Fe_3O_4, (e) Pd spherical NPs/C@Fe_3O_4 and (f) Pd cubical NPs/C@Fe_3O_4. Figure adapted from the Sciencedirect [175].

activity of the nanocatalyst was linked to the shape effect of the Pd cNPs. Despite their larger particle size (13 nm) of cube shape NPs than that of the spherical Pd NPs (3 to 4 nm), due to the [96] facet present on the cube shape Pd nanoparticles (cNPs), they were highly active for the Heck coupling reaction. That indicated that controlling the shape of the NPs to have more active facets on the exposed surface is important in optimizing catalyst performance.

For the enhanced catalytic activity and recoverability of the nanocatalysts, stabilizing catalytic active nanocatalysts onto the MNPs is important. Further, to avoid the agglomeration of the NPs, the Sekar group synthesized the Pd NPs stabilized with organic corona and employed them as reusable catalysts for C — C bond-forming reactions (Scheme 14) [176]. The nanocatalyst exhibited high catalytic activity, providing the corresponding biphenyls and diphenylethene derivatives in good to excellent yields (up to 96 %). Besides, the nanocatalyst could be quantitatively recovered and reused without appreciable catalytic activity loss. Due to the higher Pd-C bond strength, the enhanced catalytic activity was linked to the high stabilization of the Pd NPs with organic shells. This, in turn, helps in restricting the agglomeration of the NPs and maintaining the intact particle size. Antonio Monopoli and his co-workers reported the electrochemical impregnation of Pd NPs over tetragonal zirconium oxide nanopowders. The electrochemically impregnated Pd/ZrO_2 was demonstrated as an efficient metal-based nanocatalyst for coupling reactions (Suzuki and Heck) of aryl halides in an aqueous environment. While during the coupling reactions, the phase transfer catalyst and base tetrabutylammonium hydroxide (TBAOH) produced an emulsified reaction mixture, which overcame the solubility issues of low polar materials in water conditions [177].

The 2D graphene oxide (GO) structures, being rich in oxygen functional groups and having a large specific surface area, qualify as robust support for a series of heterogeneous nanocatalysts [178,179]. Parmanand et al. reported the synthesis of heterogeneous nanocatalysts (GO-Pd) by immobilizing Pd NPs onto Schiff base-functionalized GO. The GO-Pd was utilized as a nanocatalyst for Suzuki and Heck coupling reactions [180]. The nanocatalyst exhibited promising catalytic activity by providing the corresponding biphenyl and diphenylethene products with yield (up to 95 %). The catalyst could be easily recovered from the reaction without showing any significant change in the surface morphology as monitored by SEM analysis. Moreover, the recovered catalyst could be reused up to five times without any significant loss of catalytic activity [180]. More recently, our group reported the easiest synthesis approach for the immobilized Pd NPs over highly reduced graphene oxide and used for Heck coupling reaction in water. During the synthesis of Pd-based nanocatalyst, amino pyrene was used as functionalizing ligand, which provides chemically active binding



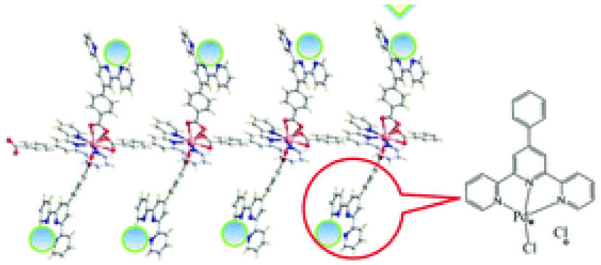
Scheme 14. Synthesis of Pd NPs, application for the Heck reaction.

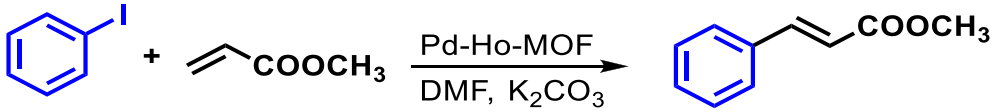
sites for the homogeneous and effective nucleation of Pd NPs on the surface of highly reduced graphene, as a result, it enhanced dispersibility and physical stability of HRG-Py-Pd nanocatalyst during Heck coupling reaction [181].

Dong and co-workers used a holmium MOF for the Heck and Suzuki coupling reactions. [Ho(2-TriPP-COO)$_3$] was synthesized by the treatment of holmium nitrate with 4′-(4-car-boxyphenyl)-2,2′:6′,2′′-terpyridine [182]. The MOF was functionalized with $PdCl_2(CH_3CN)_2$. The parent MOF crystallized in a monoclinic space group with an anion-coordinate HoIIInode linked to six oxygen atoms and three nitrogen atoms. The SBU had a distorted tricapped trigonal prismatic geometry (Scheme 15). The MOF was then utilized for the Heck coupling reaction of iodobenzene with methyl acrylate. As expected, aryl chlorides produced no discernible conversion. Interestingly, even aryl chlorides produced excellent conversions. Pd leaching was found to be 86 ppb [182].

In the Sonogashira reaction, the cross-coupling of aryl or vinyl halides with terminal alkynes to produce conjugated enynes and aryl alkynes is also catalyzed by Pd, using copper (I) as a co-catalyst. The reaction finds many applications in synthesizing natural products, pharmaceuticals, and other important molecular frameworks [183–190]. Although Pd-catalyzed Sonogashira reaction- under homogenous conditions that employ complexes of Pd(II) salts and phosphine ligands display high activity, it is quite challenging to recover from reaction mixtures [187]. This makes its recycling complicated for reuse [187]. Therefore, the heterogeneous Pd nanocatalysts, consisting of an active catalytic Pd species supported on solid support, have been a focus of recent research interest from the scientific community [114–120].

Polanski et al. developed a novel methodology for a copper-supported Pd nanocatalyst by transferring nanoparticles to the target Cu by intermediate carrier SiO_2 [188]. The as-prepared

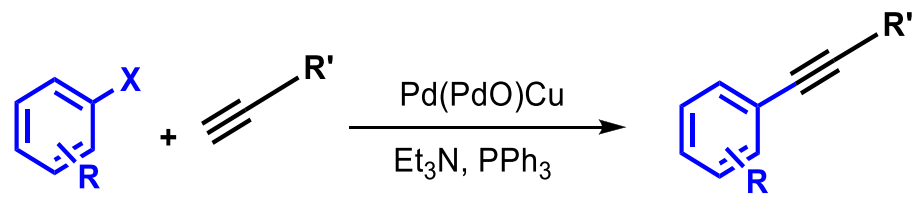




Scheme 15. Heck coupling reaction catalyzed by Pd-Ho-MOF Reproduced from Ref. 186 with permission from the Royal Society of Chemistry.

bimetallic Pd/PdO/Cu nanocatalyst was then used for a series of Sonogashira reactions. The Pd/PdO/Cu was a highly efficient and selective catalyst for the reaction between aryl halides and various terminal acetylenes, giving yields from 63 to 100 % of the corresponding products (Scheme 16) [188]. Ultrasonication was employed to prevent the aggregation of NPs and retain the catalytic activity. Moreover, the recycling of catalysts was facilitated through catalyst separation from the reaction mixture using centrifugation. In addition, a known amount of phosphine ligand was used in each round of the cycle to stabilize and inhibit catalyst deactivation. The performance of the nanocatalysts was proved to be unchanged within the two repeated cycles (100 % conversions). The role of Cu was regarded as two folds; (i) as solid support and (ii) as co-catalyst. The high catalytic activity of the bimetallic nanocatalysts was linked to the coexistence of PdO species, which was developed in the Pd/Cu system as revealed by X-ray photoelectron spectroscopy (XPS) analysis [188].

The significant development concerned the elimination of Cu (I) species used as co-catalyst since it can persuade homocoupling reaction of the terminal alkynes to dienes with oxygen (Glaser-type reaction) [189]. In 2005 Astruc et al., summarized the research work in the field of Pd-based nanocatalysts for coupling reactions under ambient conditions. He concluded that the low generation of metallodendrimers or metallosters usually optimizes the efficiency and recyclability of catalysts [190]. Later in 2007, they reviewed Pd NPs as catalysts in C—C coupling reaction as homogeneous or heterogeneous pre-catalysts and whether these NPs are intentionally performed or produced from Pd derivatives such as Pd $(OAc)_2$. Among the coupling reactions, the most studied is the Heck coupling reaction of halogenoarenes with olefins, usually using high temperatures (between 120 and 160 °C). Under this harsh condition, the precursor Pd^{II} was reduced into Pd^{o}, generating Pd NPs from which generally Pd atoms leach, which subsequently oxidizes aryl halide onto the surface of Pd NPs as a source of molecular catalysts [191]. In the same year, Astruc and coworker investigated the role of loading a low amount of metal nanocatalyst for developing Pd-catalyzed C—C coupling reactions such as Suzuki and Heck reactions. They reported that dendrimer stabilized PdNPs (1 ppm) catalyze Suzuki reaction under ambient conditions [192]. Besides, the lower amount of catalyst loading is crucial for the development of green Pd-assisted nanocatalysts to catalyze C—C cross-coupling reactions.[7] Seminal studies from the groups of Beletskaya, Reetz, and de Varies showed the Pd NPs can catalyze Heck and Suzuki-Miyaura reactions with aryl iodide and, in some cases, aryl bromides at parts per million levels [7,193,194]. As a result, researchers coined the term "homeopathic" Pd catalysis. The catalytic efficiencies achieved "homeopathic" (0.1 and 0.0001 mol percent catalyst loading achieved TON = 660 and TON = 540 000 respectively) behavior matter the PdNP size and stabilization type. The TON increased with decreasing the Pd/substrate ratio, which suggested a leaching mechanism. [195] According to the atom leaching mechanism dendrimer stabilized PdNPs synthesized by using click chemistry approach catalyzed C—C cross-coupling reactions under ambient conditions with the homeopathic amount (little as 1 ppm) of PdNPs. It is also investigated that the extraction of these PdNPs from reaction mixture, which is a complicated task in synthetic chemistry, is possible with the thiols group as catalytically active PdNPs of complementary mechanistic interest. Therefore, it is also feasible to determine the influence of NP size on the catalytic parameters (efficiency and stability) and mechanism [192,196]. Recently, the Ali group reported that heterogeneous Pd-based nano catalytic systems (Fe_3O_4/P_4VP-Pd) are recommended as an efficient catalyst for C—C coupling reactions. Due to its high reaction yield (98 %), convenient synthetic process, good biocompatibility of the catalyst, reusability, and economic viability, the Fe_3O_4/P_4VP-Pd could be suitable for commercial or industrial applications [197]. Likewise, in another important report, Wei et al. investigated the role of immobilized Pd NPs over pectin-coated magnetic iron oxide (Fe_3O_4) nanomaterials (Fe_3O_4@pectin/Pd). The Pd-based Fe_3O_4@-pectin/Pd nanocatalyst showed excellent reactivity in the C—C coupling reaction when an array of various biphenyls and aryl amines were procured by the reaction of aryl halides with phenyl benzoic acid. The catalyst was easily recovered using a magnet and recycled with insignificant Pd leaching or loss in overall catalytic performance [198]. In 2013, the Astruc group reported an interesting study using mono and polymetallic Pd-based nanocatalysts for C—C coupling reactions. He and his group synthesized mono and polymetallic Pd complexes having 2-pyridyl-1,2,3-triazole (pyta) and nano branched (nonapyta) ligand using palladium acetate and these ligands by 1:1 ratio and applied as catalysts for C—C coupling reactions such as Suzuki, Heck, and Sonogashira coupling reactions. Besides other factors, these catalysts prioritize their robustness, allowing reusability at least up to times [199]. Pd NPs incorporated in MIL-101(Cr) [Pd/MIL-101(Cr)] were synthesized and utilized for Sonogashira cross-coupling reaction of aryl and heteroaryl bromides with various alkyne in the absence of copper. The average particle size of Pd in Pd/MIL-101(Cr) was 3–4 nm, as evidenced by TEM analysis. Under the optimized reaction conditions, Pd/MIL-101(Cr) afforded 90 % yield of 4-nitro-1,2-diphenylacetylene at 130 °C using CH_3COOK as a base. Pd/MIL-101(Cr) promoted the formation of a series of 1,2-diphenylacetylene derivatives in moderate to high yields. The stability of Pd/MIL-101 was proved by reusing it in four consecutive cycles without any significant loss in its activity. Its may be due to the high surface area, large pore size and stability to air and moisture the Pd/MIL-101 acts as highly active catalyst without any additives.



R = H, NH_2, CHO
R' = $Si(CH_3)_3$, CH_2OH, Ph
X= Br, I

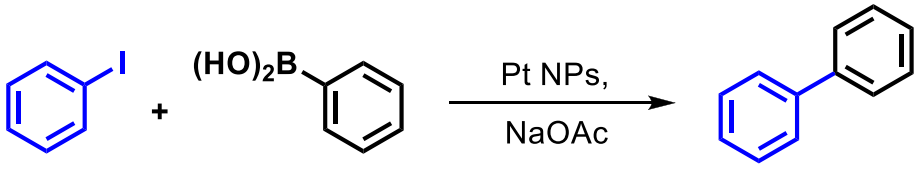
Scheme 16. Pd/Cu-based catalyst for the Sonogashira coupling reactions.

3.2. Platinum nanocatalyst assisted C—C bond-forming reactions

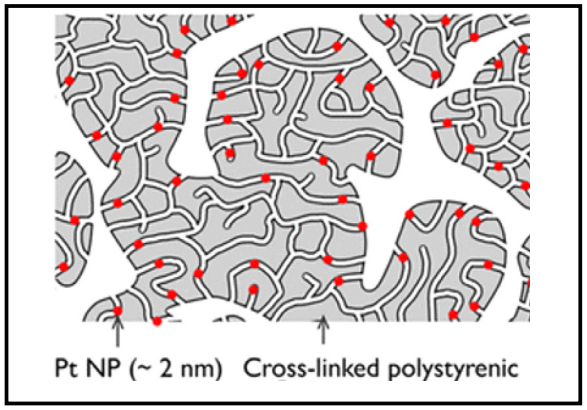
Platinum as nanocatalysts for cross-coupling reaction has been utilized to a limited degree compared to Pd NPs because the oxidative and reductive elimination steps are slower than those of Pd NPs. However, Narayanan et al. have synthesized tetrahedral-shaped poly(vinylpyrrolidone) stabilized Pt NPs to catalyze the Suzuki coupling reaction of aryl halide-like iodobenzene and phenylboronic acid (Scheme 17) [66]. They concluded that the morphology of the Pt NPs could be altered from spherical to tetrahedral shape through tailoring reaction parameters which, in return, greatly affect the progress of the Suzuki coupling reaction [200]. In comparison, spherical shape Pt NPs were found to be more active than tetrahedral shapes. However, the authors further investigated that the Pt NPs are not chemically stable under reaction conditions. This leads to a rapid decrease in the catalytic activity of the nanocatalyst.

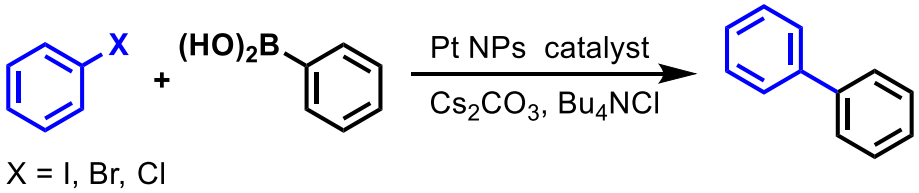
Specific engineering of nanomaterials shape, size, surface composition, and adsorbed capping ligand is highly important to control selectivity and catalytic activity. This also differentiates the colloidally synthesized Pt-based nano-catalysts from normally used heterogeneous catalysts [201]. Kim et al. developed a block copolymer with tunable micro and mesoporous structure loaded with 2 nm size Pt NPs as a nanocatalyst and utilized in ligand-free Suzuki cross-coupling reaction (Scheme 18) [202]. The higher catalytic reactivity and efficiency of Pt-HPP (Hierarchically porous polymer) to a challenging substrate like aryl halide is striking, especially in aryl chloride, where the stronger C-Cl bond disfavor oxidative addition reaction. The screening of the prescribed substrate scope revealed Pt-HPP mostly works well, but electron-deficient substrates lowered the desired yield of selective product. This Pt-loaded polymer support (Pt-HPP) outperformed mesoporous/microporous based supports and indicated higher TOF and better reusability, confirming the hierarchically porous nature of Pt-HPP. It also revealed the superior performance contrasted with analogs of mesoporous silica such as MCF-17 and SBA-15 in terms of TOF, compatibility to green solvents, and especially the substrate's scope. The hierarchically porous block copolymer has microspores within a 3D continuous mesoporous wall, which was then produced by combining the polymerization-induced micro-phase separation process [203] with hyper cross-linking. The reactant substrate rapidly diffuses through the hierarchically porous structure and accesses the large surface provided by the microspores, where the NPs are anchored. The robust hyper-cross-linked framework offers high chemical and thermal stability. Further, the catalytic activity of the prepared catalyst was found promising for the Suzuki coupling reaction under mild conditions and ensuring recyclability [204].

Corma synthesized platinum clusters with <5 atoms and used them in base-controlled Suzuki, Heck, and Sonogashira coupling reactions [205]. This computationally explained the base-controlling effect that rolls out from overcoming the energetic barrier of the oxidative addition and translating the rate-determining of the coupling to alter the step where the base actively contributes [205]. For the Heck C — C coupling reactions, Pd and Ni-based catalysts are more frequently reported in the literature. While the other heavier members of its group, such as Pt, barely perform in the same manner. Therefore, there are only some salts or com-



Scheme 17. Pt-based nano-catalysts catalyzing the Suzuki coupling reaction.





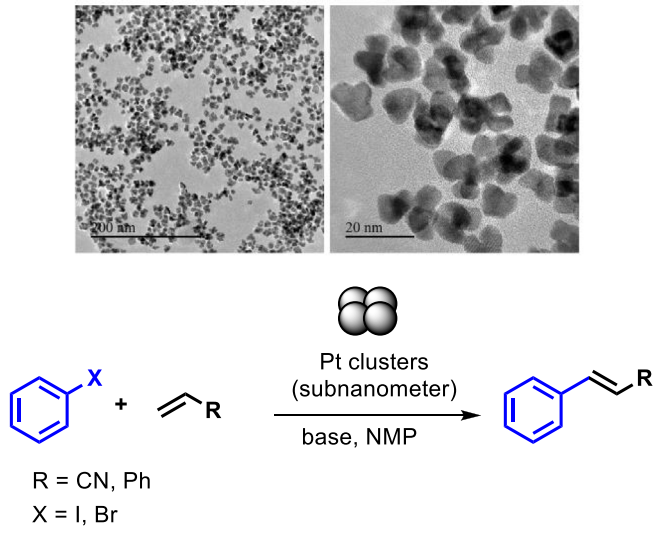
Scheme 18. Schematic illustration of the Pt NPs on porous materials for the Suzuki coupling reaction, along with substrate scope [202].

plexes of Pt, catalytically active for Heck reaction due to its dissolution in organic solvents such as dimethylformamide, etc. However, Estefania et al. recently reported Pt NPs as very active species, and these clusters efficiently catalyzed the Heck C—C coupling reaction. The author also differentiated the barriers in the first elementary mechanistic steps with small nano-size catalysts (Scheme 19) [205]. These are conventionally found for organometallic compounds and, with the oxidative addition step not the rate-determining step of the reaction. This may be due to the metallic nature of active sites and the participation of bimetallic or trimetallic atoms in the process [13]. It also makes easier cleavage of the C – X bond of iodo and/or bromobenzene, which allows the higher stabilization of fragments for the required coupling reaction. Consequently, higher activation barriers are required for the coupling step of alkene. Nevertheless, the insertion of only one molecule of each suitable reactant neglects the consequences of co-adsorption of other solvent molecules or reactants was noticed. Certainly, it is precise that the adsorbed solvent molecules may favour the recombination step, which may delay the bond-breaking step in the reactions [205].

3.3. Iron-based nanocatalyst assisted C—C bond-forming reactions

Recently, the use of iron (Fe) based catalysts in organic transformations, particularly in C—C bond-forming reactions, received significant attention due to their excellent catalytic and magnetic properties [206–215]. In this contribution, iron, as zerovalent or multivalent, is used either as a catalyst or as magnetic support material [209–218]. As an example, Akiyama et al. used sulfur-coated Au-supported iron (0) (SAFe(0)) nanoparticles as ligand-free nanocatalysts for Suzuki coupling and C—H amination reaction [212]. Detailed spectroscopic analyses showed that the iron (0) NPs with a size of 6–10 nm, which tend to be easily oxidized to iron (III), were stable in air when immobilized in SAFe(0) with self-assembled multilayers. The catalyst was simple to treat in the air and could be reused in reaction processes with a low iron leaching rate showing a great catalytic activity for forming a C—C bond in an excellent yield of 67–95 % (Scheme 20).

Additionally, the assorted potentials of Fe as a result of convenient oxidation and spin states could be extended by the designed and engineered ligands, which interfere efficiently with the active



Scheme 19. Pt nanoclusters catalyzed Heck reaction. High-resolution transmission electron image of various Pt NPs used as a catalyst for the Heck coupling reaction [205].

catalytic processes [213]. Precious metals like Pd, Pt, and Au, supported on ferrimagnetic NPs have a diverse range of catalytic applications, especially in C—C coupling reactions such as Pd NPs supported on nanoscale bio-magnetite effective catalytic activity in Heck coupling reaction [214]. Considering this prospect, the research community predicted that in the future, several novel Fe-based catalyzed reactions and their potential application in organic synthesis will be inevitable [215,216].

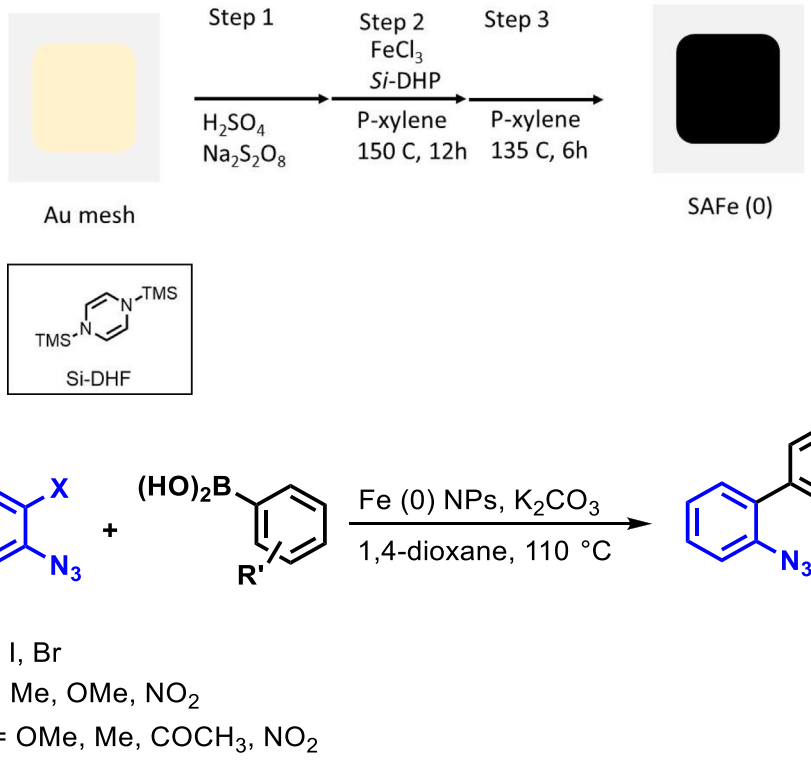
Buchwald and Bolm et al. highlighted a fundamental and exciting question about the activity of metal impurities in $FeCl_3$ when it is applied as a catalyst for C—C bond formation [217]. They have explained in their discussion that including a trace quantity of copper oxide in the reaction mixture has enhanced the cross-coupling reaction [218]. The Sharma group also reported the magnetically efficient and retrievable nano-catalytic system (Cu-2QC@AM-SiO_2@Fe_3O_4) with intrinsic stability in solvent-free conditions [219]. As an activated nucleophile, a significant enhancement in the catalytic activity of the Fe-catalyzed Suzuki coupling reaction between alkyl halides and lithium aryl boronates has been reported. For example, the research groups of Hatakeyama and Nakamura used Fe diphosphine as catalysts for the different cross-coupling reactions, including Negishi, to synthesize olefins [220–229]. Zhou decorated Pd on Fe@Fe_xO_x and then utilized it as a catalyst for the Suzuki cross-coupling reactions in an aqueous medium at room temperature under the normal lab environment (Scheme 21) [223]. He coupled the profits of heterogeneous (such as stability in the air, low cost, easy separation, and excellent reusability) and homogeneous systems (such as high yield), which fabricated it a promising catalyst for practical application [224].

Marandi et al. recently reported a core–shell nanostructured magnetic nanocatalyst for Heck coupling reaction by immobilizing Pd (II) onto triethanolamine (TEA) functionalized Fe_3O_4 nanoparticles. This cost-effective catalyst showed significant reactivity, efficiency, and stability toward the C — C coupling reaction of olefins and other aryl halides (Scheme 22). This catalyst's other advantages include facile reaction conditions, an easy workup method, and minimal Pd leaching. Also, this catalyst is easily dispersed in the reaction medium, reducing the reaction time. Therefore the catalyst could be used for other diverse reactions as a promising candidate in the future [225].

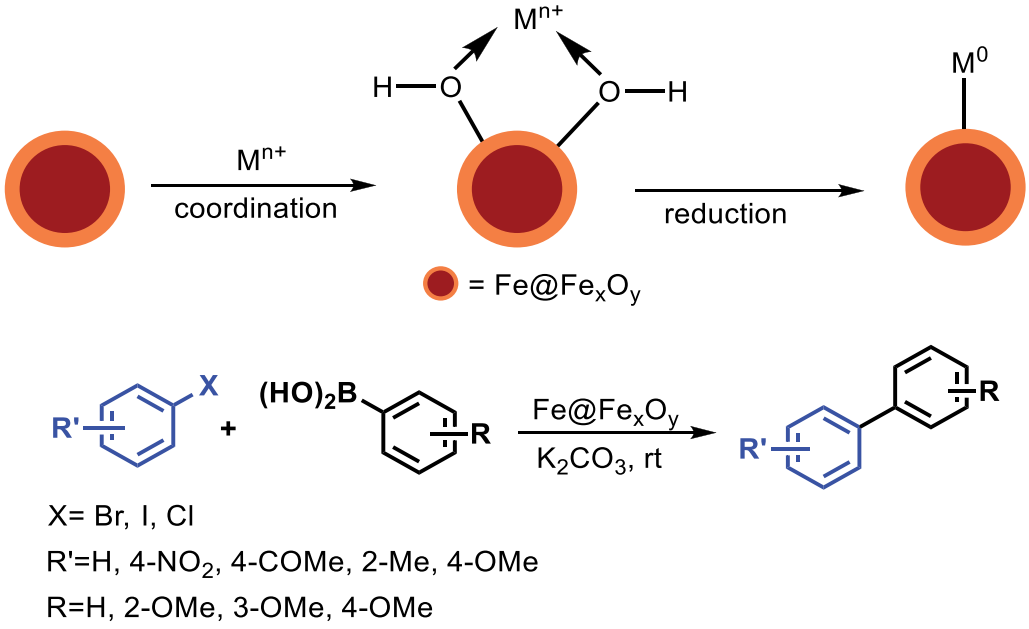
To expand the substituent scope of the substrates, Nakamura and his co-workers conducted a series of alkyl halides coupling reactions with magnesium tetraalkylborates obtained under the optimized conditions from olefins, BH_3-SMe_2 and iPrMgCl. They prove that this reaction was not only applicable to primary alkyl halides but also a valid option for secondary alkyl halides having bromide functional groups. They investigated that the loss of stereochemistry in secondary alkyl halides having bromide functional groups is due to the radical process during reaction. It is noted that alkane production is a major side product during the reduction of alkyl bromides in a moderate yield. The 1-Bromo-4-(2-bromoethyl)benzene experienced coupling reaction with tetraalkylborate via selective cleavage of C_{sp3}Br bond. But however, cleavage of C_{sp2}Br bond was not observed during the reaction. Thus, this selective cleavage of C_{sp3}Br bond could be credited to the radical character of organoiron species and as well as to the stabilities of the corresponding radical intermediates [222].

3.4. Copper as nanocatalyst for C—C bond-forming reactions

Transition metal-catalyzed cross-coupling reaction to generate a C—C bond has revolutionized chemical science [13,226,227]. Despite significant accomplishments, these metal catalysts also raise some major issues, including their high price, need for certain ligands, and sensitivity to air. However, to solve these issues, var-



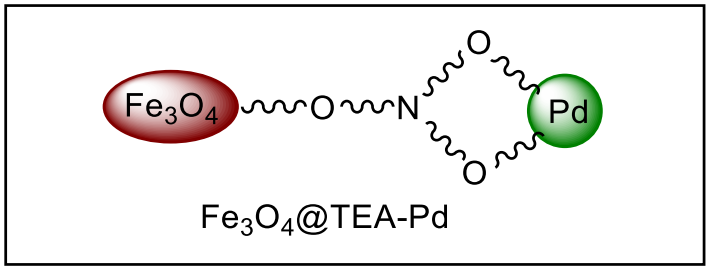
Scheme 20. Liquid phase combinatorial synthesis using SAFe (0) and aryl iodide.

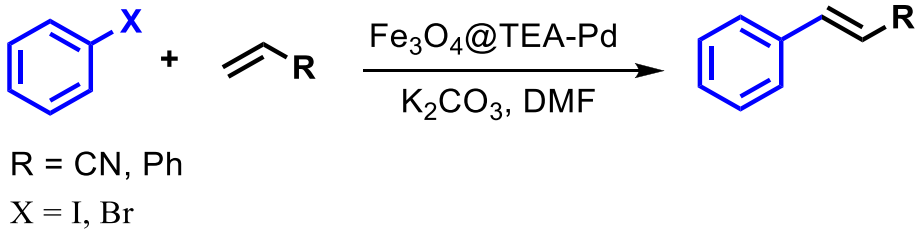


Scheme 21. Mechanistic illustration for the deposition of metallic domains on the surface of $Fe@Fe_xO_x$ and its utilization for the Suzuki coupling reaction.

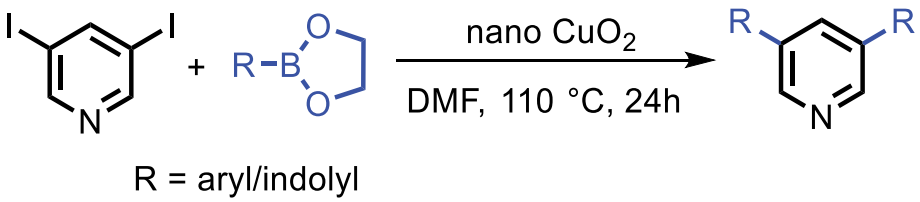
ious techniques are reported for synthesizing cheap and abundant available metals such as copper (Cu) [228]. The Cu nanocatalysts are prepared by various methods such as hydrothermal [229], co-precipitations [230], homogenous precipitation [231], and impregnation [232]. Ranjani et al. reported the *in-situ* generation of Cu_2O NPs using $CuCl_2$ as a copper precursor and K_3PO_4 as a reducing agent in DMF. This nanocatalyst was used for the ligand-free catalyzed Suzuki coupling reaction of 3,5-diiodopyridine with aryl and indole boronates. The reaction resulted in an excellent yield (Scheme 23) [233]. [233]. Cu-assisted nanocatalysts hold significant interest due to their low reaction temperature and high catalytic activities [228]. The catalytic activity could enhance by altering temperature, concentration, pressure, reaction type, and synthetic protocols [234].

Recently Lamei et al. reported the ligand-free, non-toxic, green catalytic materials consisting of Cu nanowires and NPs synthesized on an etched carbonaceous matrix for Suzuki coupling reactions (Scheme 24). Such Cu-based nanocatalyst showed excellent activ-

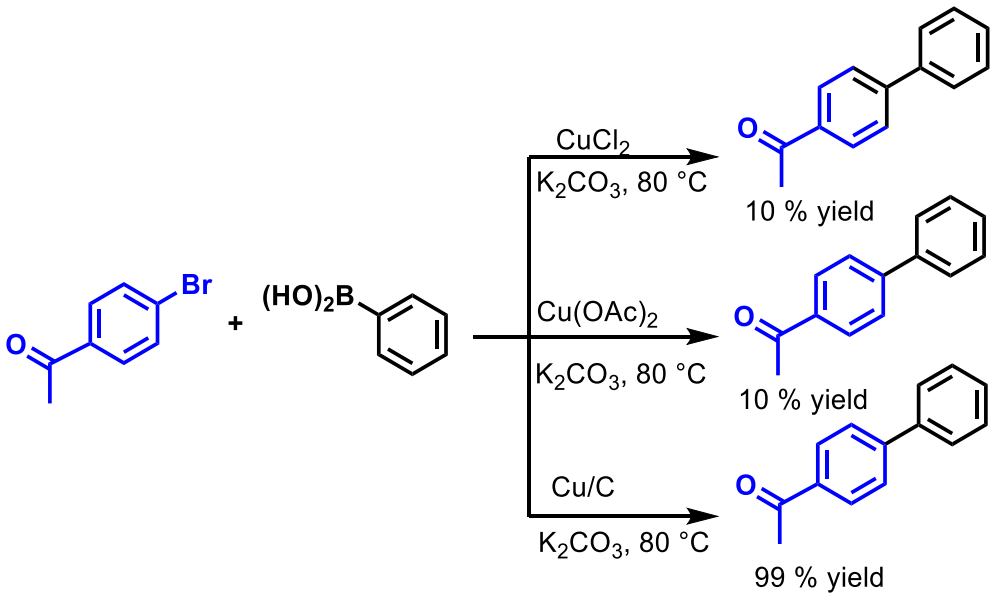




Scheme 22. Fe_3O_4@TEA-Pd catalyzed Heck coupling reaction.



Scheme 23. Nanosized CuO_2 catalyzed Suzuki coupling reaction.

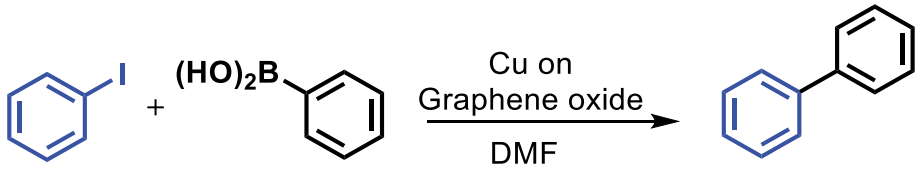


Scheme 24. Various Cu-based catalysts for Suzuki coupling reaction and general cycle for coupling reactions.

ity (99 % yield) with no significant activity loss even up to four cycles [235].

Shabestari et al. reported for the first time on the facile decoration of graphene oxide materials with different copper-based nanostructures using microwave-assisted methods [236]. Further, the Cu-based graphene catalyst was utilized for the coupling reaction of iodobenzene and phenylboronic acid at 110 °C for 24 h, while 83 % conversion was achieved (Scheme 25). The main reason for the Cu-based graphene's high performance is the catalyst's higher specific area, better dispersion, and the presence of hydroxyl group, which may further increase the basicity of the catalyst helping in the coupling reactions.

Copper terephthalate [Cu(benzenedicarbocylate)] (benzenedicarboxylate = bdc) MOF has been reported as a recy-

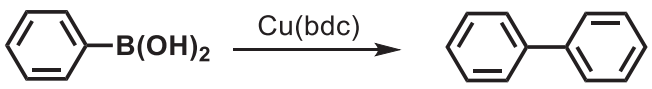


Scheme 25. Cu-based graphene catalyst for the C—C coupling reaction.

clable catalyst for the aerobic homocoupling of arylboronic acids to the corresponding symmetrical biphenyls under mild reaction conditions [237]. The reaction was optimized with phenylboronic acids as a model substrate, achieving a maximum yield of 97 %in DMF at room temperature (Scheme 26). The heterogeneity test showed the absence of metal leaching under experimental conditions. The catalyst was reused five times with a slight decrease in the yield of biphenyl. It is concluded that Cu(bdc) exhibits higher activity over other conventional catalysts such as Cu, Pd, Fe, and Au, but this conclusion has to be taken cautiously when the activity of Cu(bdc) has been measured under different conditions than those employed with other catalysts.

In the other study, Cu_2O NPs supported on rGO, synthesized using a simple solution-phase reduction approach, and employed efficiently to catalyze the Sonogashira cross-coupling reactions [238]. The catalyst Cu_2O/rGO nanocomposites exhibit high selectivity and catalytic activity towards the coupling reaction of aryl halides with phenylacetylene in ligand-free conditions. In the catalytic mechanisms, Cu_2O NPs first activated the phenylacetylene to produce copper (I) acetylide compounds in the presence of a base. These compounds reacted with aryl halides to form diphenyl ethylene (Scheme 27). After five cycles, the catalytic activity of the reaction slightly decreased due to leaching and partial oxidation of Cu_2O in strongly basic conditions. The Cu_2O/rGO nanocomposites heterogeneous catalyst is a good option to provide future direction for the replacement of palladium-based catalyst for the Sonogashira cross-coupling reactions [238].

In other studies, Zhao proposed Cu_2O NPs as a catalyst for coupling reactions such as Sonogashira between terminal alkynes and aryl iodides in DMSO as solvent at a temperature of 160 °C under an argon atmosphere in the presence of K_2CO_3 as a base [239]. The coupling reactions between aryl halides and terminal alkynes have been carried out in the presence of Cu NPs using an organic base (Scheme 28) [240]. Phenylacetylene reacts with a Cu cluster, forming a [alkenyl-Cu] cluster, which further reacts with the aryl halide and finally leads to the product by reductive elimination steps [241]. Notably, the positive charge generated during the oxidative addition is shared between the Cu atoms in the cluster, which is advantageous in contrast to the monoatomic copper complexes. This assists product formation and the reductive elimination steps in the reaction [241].



Scheme 26. Aerobic oxidative homocoupling of phenylboronic acid by Cu(bdc) MOF.

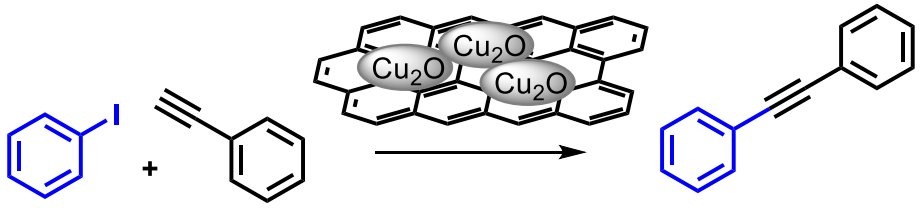
Park et al. synthesized Cu_2O NPs supported onto the mesoporous silica (Cu_2O/SiO_2) as a novel eggshell shape catalyst using a simple impregnation strategy [242]. The general synthetic approach for the preparation of Cu_2O/SiO_2 egg-shell nanosize catalyst is shown in Fig. 9.

Initially, a silica eggshell was fabricated using a slightly modified sol–gel method. The supported catalyst with narrow pore size distribution and a large surface area showed high catalytic activity towards the Sonogashira reaction of acyl chlorides with terminal alkyne (Scheme 29). The high activity is imputed to the strong synergizing effect between highly dispersed Cu_2O NPs and mesoporous silica [242].

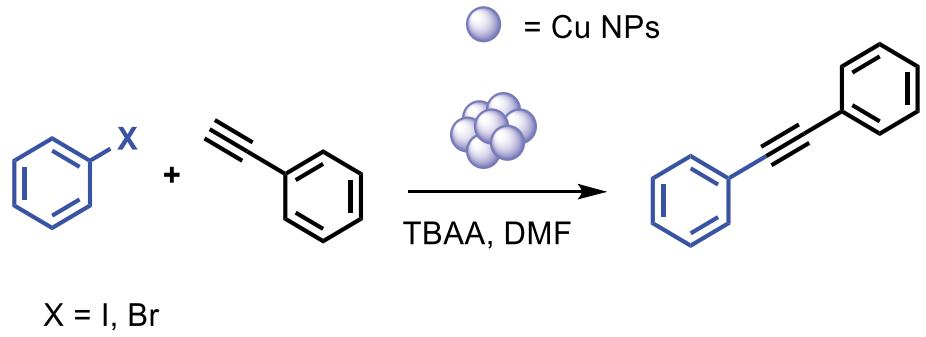
The Sonogashira coupling reaction of phenylacetylene and iodobenzene was catalyzed by (1,4-diazabicyclo[2.2.2]octane) DABCO-based Cu(I)-MOF catalyst, and afforded the corresponding product in 76 % yield under an N_2 atmosphere [243]. Recycled catalysts exhibited consistent activity even after three cycles. To further explore the catalytic activity of Cu-based nanocatalysts for various coupling reactions of aryl halides with phenylboronic acid have been utilized under optimized conditions. They observed that electron-withdrawing groups, such as substituted halobenzene and its derivatives, were transferred into their corresponding products, such as biphenyls, with an excellent yield. Likewise, phenylboronic acid substituted with electron-donating groups with halobenzene and 3-nitrohalobenzene were reacted and yielded the corresponding biphenyl with an excellent yield. Optimizing the reaction's catalyst loading and time intervals was the most challenging task. For example, chlorinated substrates needed high mole percent of the catalyst loading (0.2 %) with time interval of 4.4.5 h. The electron-rich and electron-deficient substrates led to the products with a meager yield (15–20 %) [235].

3.5. Gold nanoparticle for C—C bond-forming reactions

In the last few decades, gold (Au) has emerged as a sort of "philosopher's stone" in catalysis, being able to enhance an incomprehensible variety of organic transformations, such as C—C bond formation using cross-coupling reactions [20,252–254]. The enhancement of the catalytic activity was tailored by suitable support such as activated carbon [247], starch silica [248], transition metal oxides [249], and resin [250]. Resin-trapped Au nanocatalysts were effectively used for the Suzuki coupling reaction. Au



Scheme 27. Sonogashira reaction of iodobenzene with phenylacetylene in the Cu_2O/rGO.



Scheme 28. Cu NPs catalyzed Sonogashira coupling reaction between an aryl halide and phenylacetylene.

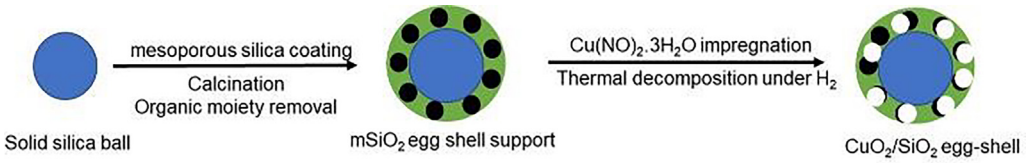
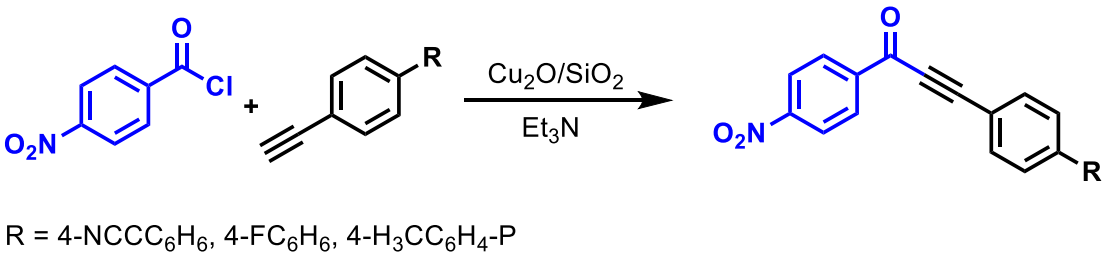


Fig. 9. Synthesis of the CuO_2/SiO_2 NPs supported on egg-shell silica [242].



Scheme 29. The Sonogashira reaction catalyzed by nano-sized Cu_2O/SiO_2 eggshell.

NPs have grown prominence in catalytic applications [251] after their use as catalysts in the facile synthesis of highly poly(2-aminothiophenol) stabilized Au NPs for the Suzuki coupling reactions [252]. The Au NPs catalyzed Suzuki coupling reaction may follow the homogeneous pathway, as the leaching of Au was observed in the initial 30 min of the coupling reactions (6 % of the initial catalyst change [253].

Recently, Haldar et al. used urea-based tryptophan as an *in-situ* stabilizing and reducing agent to synthesize AuNPs in water. The side chain of tryptophan has been applied to reduce Au ions in $HAuCl_4$ to metallic Au and the functionality of carboxylic acid to stabilize the AuNPs. Further, the stabilized AuNPs were tested to catalyze the Suzuki cross-coupling reaction among 4-bromobenzoic acid and phenylboronic acid under aqueous conditions. This resulted in a 92 % yield of coupling reaction in 40 min (Fig. 10). The excellent efficacy showed by the AuNPs catalyst was further effectively applied to many other Suzuki reactions. This result inspired the researcher in the field of catalysis to investigate coupling reactions further using AuNPs in aqueous conditions [254].

Notably, the precise control over the size and the amount of stabilizer for Au NPs are crucial parameters for enhancing their catalytic activities. As the size of Au increases from 1.0 nm to 5.0 nm, the catalytic efficiency decreases from 87 % to 10 % [252]. Conversely, controlling the size up to 1.0 nm required an excessive amount of stabilizer, reducing the reaction efficiency and producing less product yield. In conclusion, the smaller size Au NPs leads to excellent catalytic activity, whereas a higher

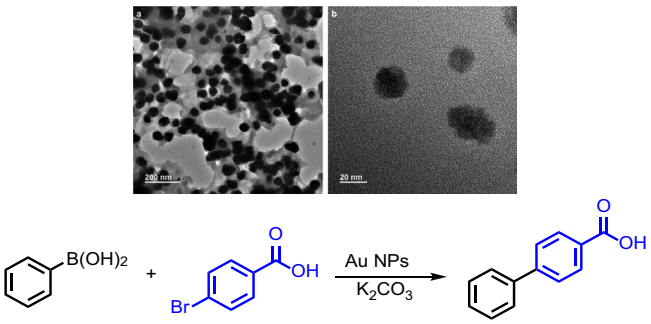


Fig. 10. Schematic presentation of AuNPs for the Suzuki cross-coupling of p-bromobenzoic acid with phenylboronic acid. (a) TEM images viewing the spherical morphology of AuNPs (b) High-resolution TEM image of AuNPs. [254].

amount of stabilizer inhibits the reaction [255]. Kang et al. reported his important contribution by introducing graphene as an interface layer for plasmonic Au NPs and active catalysts (Pd nanodots) [256]. This visible-light-driven C—C coupling reaction achieved high catalytic efficiency with Pd-rGO-AuNPs. The signifying role of the graphene interface layer is shown in (Fig. 11). These results designate the design and engineering of interfaces that are significant to obtaining the highest catalytic activity using nanocatalysts [256].

Song et al. reported water-dispersible Au@Pd core–shell and Au-Pd alloy NPs for the Heck coupling reaction [22]. In a conven-

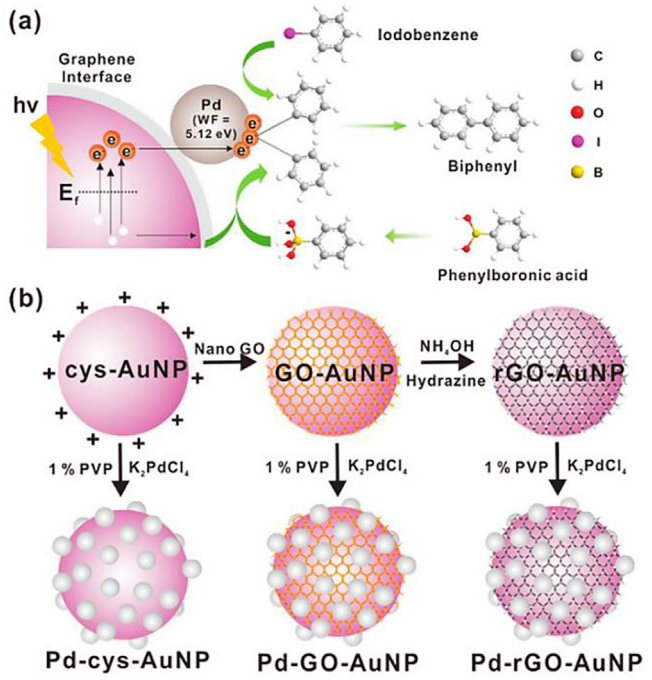


Fig. 11. (a) Schematic illustration of the Suzuki coupling reaction of Pd-rGO-AuNPs (b) synthetic scheme of Pd-cys-AuNPs, Pd-GO-AuNPs, and Pd-rGO-AuNPs [256].

tional Heck coupling reaction, Pd (0) phosphine complex system has been employed as an effective catalyst in controlling the reactivity and selectivity in the reaction, which enters the catalytic cycle of the C—C bond coupling reaction among unsaturated hydrocarbon and aryl halides. In the last step, bases were added to eliminate halides and reduce Pd(II) to Pd(0) [257]. For instance, styrene, methyl acrylate, and *n*-butyl acrylate were applied as unsaturated hydrocarbon and iodobenzene was utilized as aryl halide (Scheme 30). Au@Pd NPs presented a 99 % yield of the product after many times recycling of catalytic reactions. Cetyltrimethylammonium bromide (CTAB) was introduced as an additive to improve the activity of the catalyst. The cationic surfactant enhanced the catalytic activity effect by chelating the Pd NPs, a phosphine-free catalyst. [258] Song and co-workers utilized four different types of Au@Pd core–shell NPs at room temperature, as a result, the emulsion phase surrounded the Au core-NPs. Due to sharp branched arms, Au@Pd core–shell NPs were active in a relatively small amount throughout the whole catalytic reactions [22].

Recently Au NPs catalyst was also found to be capable of catalyzing the Sonogashira coupling between iodobenzene and phenylacetylene. Au NPs may activate the C – I bond of the iodobenzene and the C—H bond of the counter coupling partner by one gold atom at the neighbor gold atoms. [259] Corma et al. investigated theoretically and experimentally that the gold NPs are intrinsically active in catalyzing the coupling reaction between phenylacetylene and iodobenzene. [260]The author concluded that Pd is not mandatory for catalyst activity [260]. They also reported that the Au particle supported onto CeO_2 could catalyze the same reaction in the presence of K_3PO_4 as an additive without copper. The interest in Au as a heterogeneous catalyst was a key success because cationic Au species can be stabilized on the surfaces and may work as soft Lewis acids (Scheme 31) [261]. The large particles of the Au are more selective than the small ones [262].

The Sonogashira coupling reaction of aryl boronic acid and various terminal alkynes was performed readily under mild conditions. Corma et al. also showed that without an external transmetallation reagent) such as aryl boronic acid), and the homocoupling of the terminal and aryl alkynes would also proceed proficiently (Scheme 32) [263].

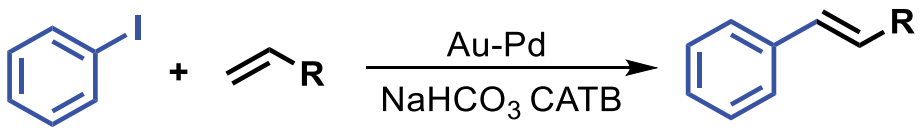
Tsukuda reported the activity of the colloidal Au NPs (<2 nm) stabilized by poly(*N*-vinyl-2-pyrrolidone) (PVP) for the coupling reaction in water under mild reaction conditions [264]. The dissolved oxygen in the water was found to be an important factor in the reaction. The experimental and theoretical investigation revealed that the stabilization of the Au NPs using PVP controls the size of NPs, resulting in enhanced catalytic activity [265,266]. Lambert et al. also reported that the atomically dispersed Au(I) and Au(II) was found catalytically inert, while the metallic Au(0) NPs were active and selective towards the Sonogashira coupling reaction of the phenylacetylene and iodobenzene (Scheme 33). The author concluded that the metallic Au is active rather than ionic Au [267].

Metal oxide-supported Au NPs have been synthesized by deposition–precipitation and incipient wetness impregnation methods [268–278]. The deposition method leads to the small Au NPs, while the wetness impregnation method gives larger and agglomerated particles [271,272]. These NPs were utilized as a catalyst for the Sonogashira coupling reaction of phenylacetylene and iodobenzene (Scheme 34). The NPs prepared by deposition and precipitation gave better conversion and selective coupling products 76 % and 75 %, respectively.

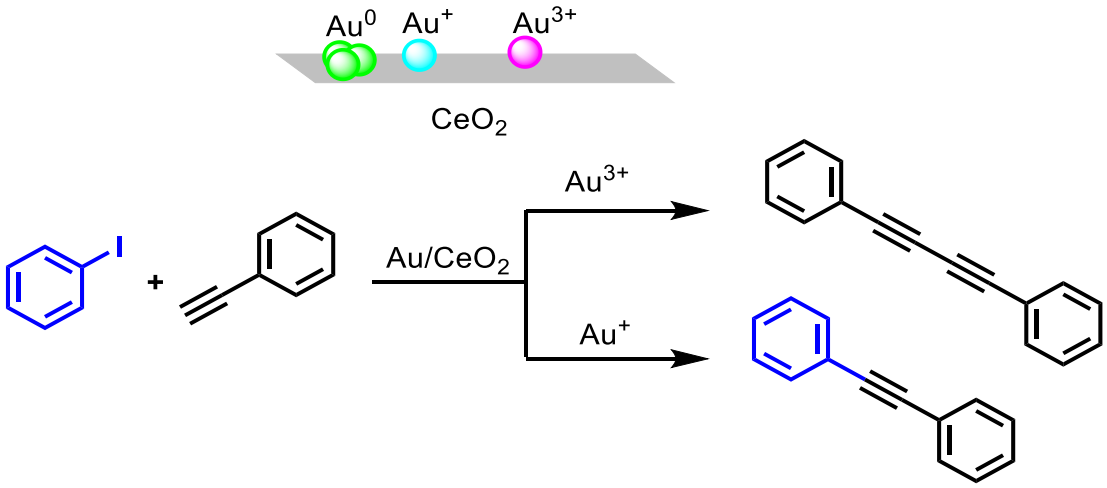
The Au NPs exhibited high efficiency in the model C—C coupling reactions. It effectively converted a large variety of functional groups, including aryl halides to 4-chloro benzoic acid, 4-chlorobenzaldehyde, and phenylboronic acid, to the corresponding coupling products in high yield. While testing further scope of the substrate, the 2-chlorobenzaldehyde showed a relatively lower yield (76 %) due to steric hindrance and electronic effects of the substrate present in the aromatic circle environment of aryl halides. In addition, it was also investigated that the reaction of phenylboronic acid with 1-chloro-4-vinylbenzene was sluggish, and after 75 min, the obtained product afforded 75 % yield [254].

3.6. Silver nanocatalyst-assisted C—C bond-forming reactions

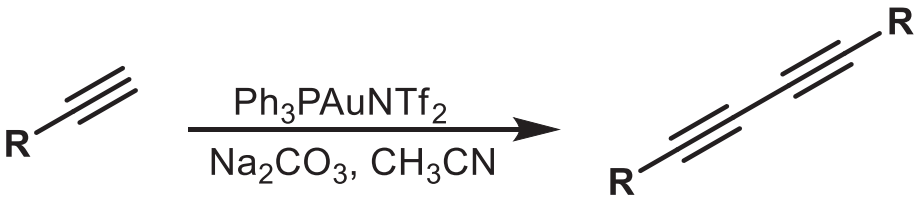
Silver (Ag) nanoparticles catalyzed organic transformations have also attracted the interest of researchers in the past ten years because of the unique catalytic properties of Ag NPs such as efficient reactivity, specific selectivity, prolonged stability, and easy recyclability during catalytic reactions [273,274]. The application of heterogeneous AgNPs in general organic transformation reactions has proven to be an effective strategy in developing cross-coupling reactions in terms of efficiency and selectivity [274]. The hybrid NPs of bimetallic Pd-Ag supported on carbon quantum dots (CQDs) is an emerging nanohybrid efficient catalyst for the Suzuki coupling reaction of aryl halides like bromides and chlorides under ligand-free ambient environments (Scheme 35). [275] The synergistic activity of the nanohybrid catalyst showed



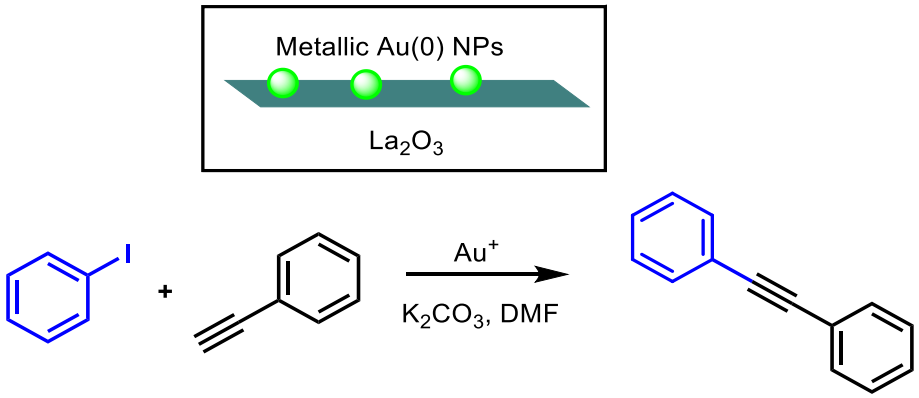
Scheme 30. Heck coupling reactions catalyzed by using AuPd alloy and Au@Pd core–shell NPs.



Scheme 31. Au/CeO_2catalyzed cross-coupling (Sonogashira) and homocoupling reactions.



Scheme 32. Nano-sized catalyzed homo-coupling reaction of the terminal alkynes.

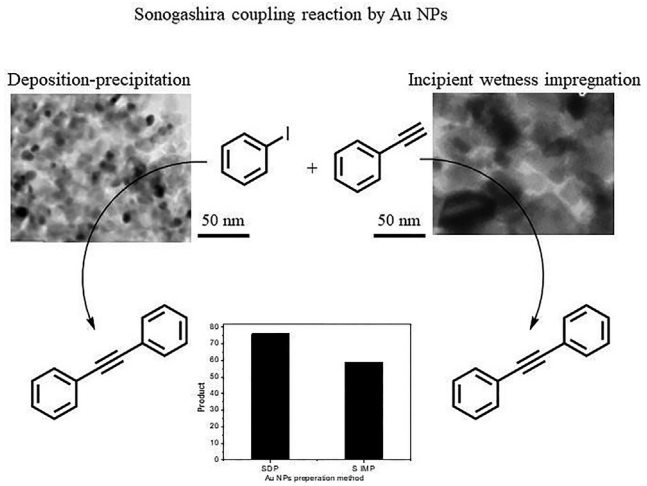


Scheme 33. Sonogashira coupling reaction catalyzed by atomically dispersed ionic and metallic Au NPs as a catalyst.

enhancement of cross-coupling reaction in a short time (<1 h) with the highest yields (>90 %). The nanohybrid catalyst showed heterogeneous character and enabled easy isolation and recyclability [275].

Recently, Chen groups reported the *in-situ* synthesized Ag NPs doped TiO_2 NPs and used for the formation of C—C bond through Suzuki-cross coupling reaction (Fig. 12) [273]. The catalytic activity of Ag NPs doped TiO_2 NPs was found to be eco-friendly and easily recyclable during cross-coupling bromoaryl with phenylboronic acid under mild conditions under visible light illumination. The doping of AgNPs boosted the catalytic activity at the surface of TiO_2 and inside the pores. The TiO_2 support provided the active sites for the accommodated AgNPs due to high surface area, therefore showing high reactivity towards the C—C coupling reaction of bromoaryl with phenylboronic acid. The higher recyclability of catalysts remarked the footprint in forming C—C coupling reactions.

Core-shell nanocomposites with CeO_2 nanoparticle shell and Ag NPs as core achieved more chemo-selectively than other metal NPs on conventional oxide support [276]. Recently, published research data showed that reduction using silver nanocatalysts, hydride, and proton polar species are designed *in-situ* on the edge of Ag NPs/hydrotalcite clay through obliging effect among the basic site of hydrotalcite clay and Ag NPs. This exclusively reduces polar functional groups. Nevertheless, the practice of hydrogen instead of CO/H_2O or alcohols in silver nanocatalysts instigates the reduction of polar groups (epoxide and nitro) and C—C bonds. The overall reaction scheme carried out, or transformation by Ag NPs is reported below in Scheme 36 [277]. In another report, the Ag NPs on basalt fibers were used as a chemical and thermal resistant surface and as a heterogeneous catalyst for Suzuki coupling reactions. The easy separation of nanocatalyst and, several times catalyst recyclability without decreasing its catalytic activity was



Scheme 34. Comparison of the Au NPs synthesis methods in Sonogashira coupling reaction [268].

economically advantageous for Suzuki coupling reactions. The proposed mechanistic pathway presented in (Scheme 35) indicated in the first step started from the oxidative addition of aryl halide, with the help of OH^- ions, and the facilitation of X^- substitution. In the second step, transmetallation was completed with the effect of boronic acid, while in the last step, through complete reductive elimination the desired final product was obtained [278].

To explore the substrate's scope, the catalytic efficiency rate of Ag NPs for the C—C coupling reactions, especially Suzuki coupling reactions, depends on the substrate substituents in the aryl halides and aryl boronic acids. The existence of electron-donating substituents in aryl halides and aryl boronic acids forces the reaction to complete in a short time, while on the other hand, the presence of electron-withdrawing groups or substituents requires a longer time for the completion of reactions. It was observed that the presence of polar nature in electron-donating moieties on the surface of catalyst facilitates better interactions and contact sites with the substituents than those of electron-withdrawing moieties [275]. In some cases, it is also possible that the transmetallation and reductive elimination occurred, but the system could not enter into a catalytic turnover [279].

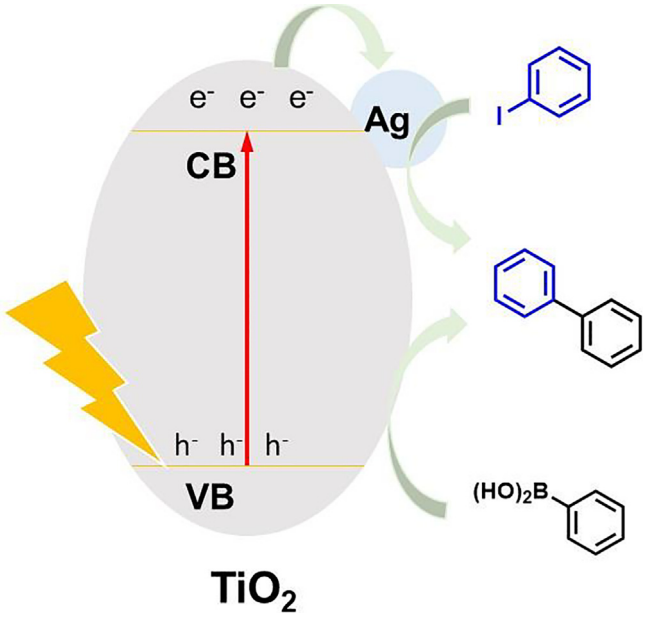
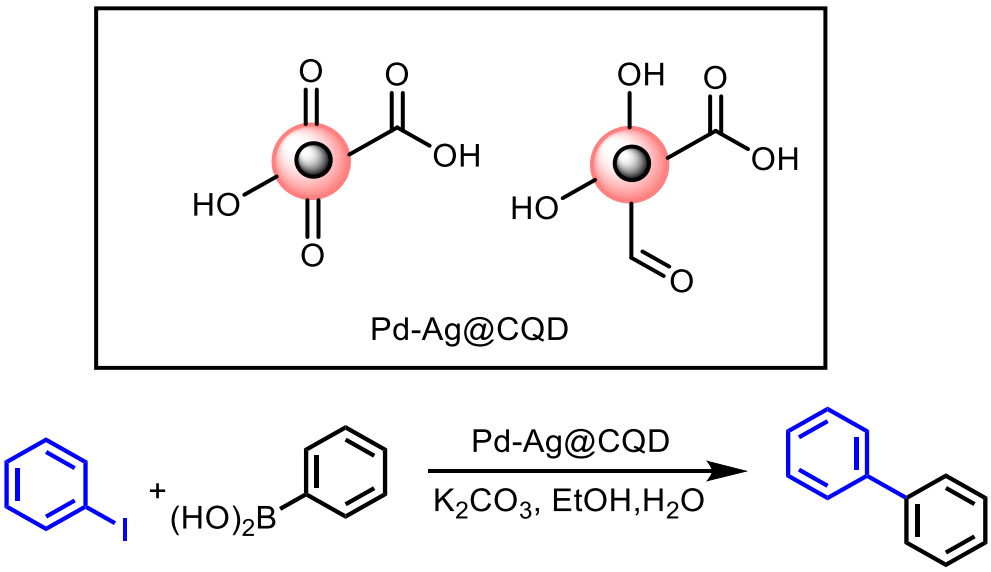


Fig. 12. Suzuki-coupling reaction catalyzed by Ag/TiO_2 photocatalyst in visible light.

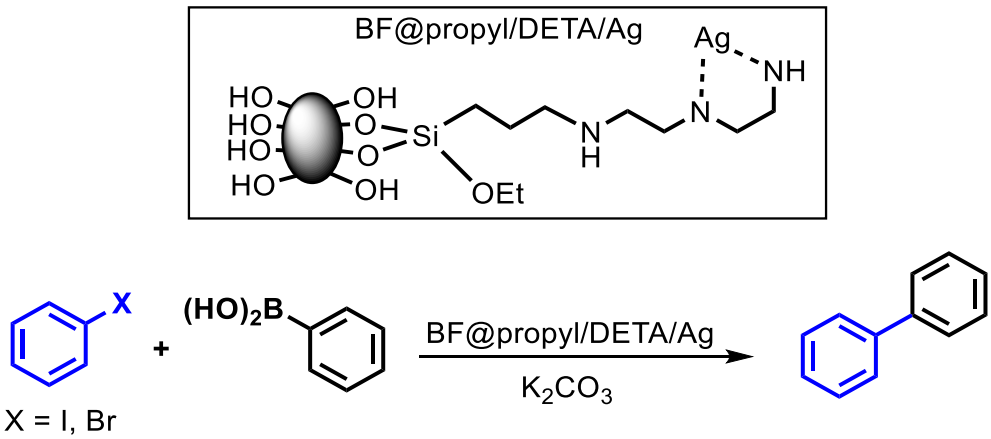
3.7. Zinc nanocatalyst-assisted C—C bond-forming reactions

Nano-sized zinc (Zn) in the composite used as a nanocatalyst for the coupling reactions consists mainly of zinc oxide nanomaterials. The ZnO NPs, when mixed with other metal oxides, exhibited good catalytic properties for Suzuki cross-coupling reactions under ligand-free and ambient conditions [280]. The synthesis of ZnO nanoparticles is accomplished using stoichiometric amounts of urea and zinc acetate in solution. Then $PdCl_2$ (3 %) was loaded onto ZnO NPs using the wet chemistry technique. The catalyst tested for and efficiently achieved a yield of 97 % for the Suzuki coupling reactions for the ring-substituted boronic acid and aryl bromides under ambient reaction conditions (Scheme 37) [280].

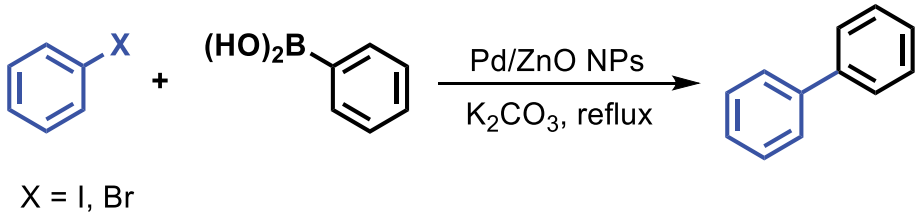
Singh et al. reported an effective superparamagnetic catalyst in the solid phase, synthesized by loading traced amount of nano-sized Pd(0) NPs during the synthesis of zinc ferrite NPs through



Scheme 35. UV-light assisted the synthesis of the Pd-Ag@CQD nanohybrid for the Suzuki coupling reaction.



Scheme 36. Suzuki coupling reaction catalyzed with BF@Propyl/DETA/Ag.



Scheme 37. Scheme of synthesized Pd/ZnO NPs using wet chemistry technique for Suzuki reaction.

ultrasound-assisted co-precipitation process and in the absence of any capping agent or surface stabilizing ligand. The Pd-$ZnFe_2O_3$ NPs showed excellent catalytic performance for both Suzuki and Heck C—C bond-forming coupling reactions under a ligand-free environment (Scheme 38). The footprint of the catalyst was its super magnetism which has been separated through a magnet and then recycled for 5 cycles without any considerable loss of catalytic activities [281].

Recently, Nouri et al. reported nanoplates of ZnO as catalysts supported by hydrazone-assisted palladacycle for the Heck coupling arylation reaction of different olefins [282]. In minutes, this ZnO-catalyzed reaction occurs under air conditions with high yields for aryl-iodides and bromides. In addition, this nanocatalyst was capable and recovered and reused for eight cycles without extensive reduction in catalytic activity. Consequently, this procedure offers an inexpensive synthetic way to synthesize stable palladacycles, which makes the ZnO nanoplates as catalyst supported by hydrazone-assisted palladacycle for the Heck coupling reaction (Scheme 39) [282].

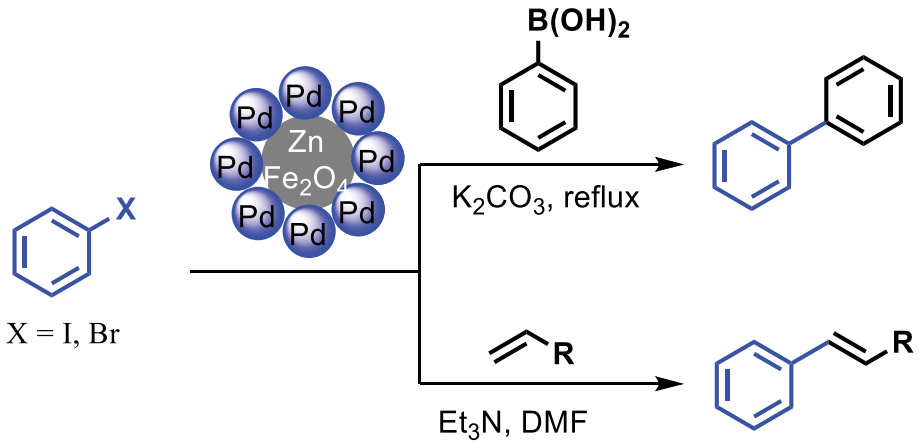
To explore the Zn catalyzed coupling reaction's generality and functional group tolerance, specifically the Suzuki reaction. Varying functionality of the substrate with electron-withdrawing and electron with donating group substituted. The protocol may work well for all the substituents, but it also depends on the substrate stability, such as electron withdrawing group (-F, -Cl) substituted boronic acid gave excellent results with 1-bromoaryl acetylenes. The methoxy substituted 1-bromo phenyl acetylene gave only moderate yields with aryl boronic acids [283]. Its due to the instability of the substrates and yielded stable aromatic compounds like acetophenone, benzaldehyde, and anisole, decreasing the formation of the desired cross-coupling products. Steric handedness also dramatically affects the reaction rate.

3.8. Nickle nanocatalyst assisted C—C bond-forming reactions

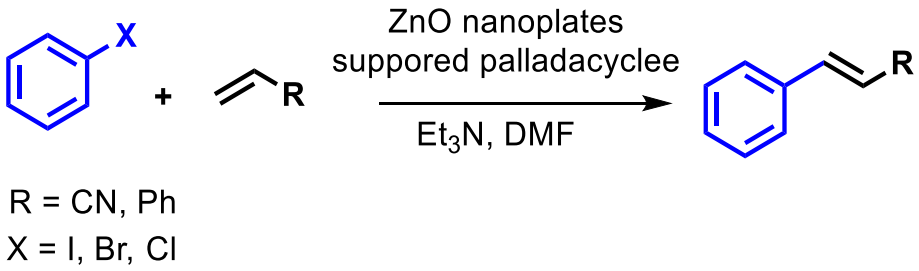
In the case of Ni NPs based nanocatalyst, although the energy barrier of oxidative addition and olefin insertion process is lower than Pd catalysts, the efficient β-hydride elimination step is more efficiently catalyzed by the Ni catalysts. [285]The stronger Ni-H bond makes the reductive eliminations step difficult and sometimes requires harsher reaction conditions [284]. Nevertheless, the deployment of low-cost transition metals such as Ni as catalysts is gaining considerable interest, and its participation in C—C bond-forming reactions [286]. Generally, the Pd-catalyzed Suzuki coupling reaction can be achieved with a broad substrate scope and high efficiency [287]. However, the reaction of economically viable chlorine substrates was slightly sluggish because of their low reactivity [288,289]. It is worth mentioning that reaction of chlorine substrates with electron-deficient aryl groups such as aryl chlorides is an option for Suzuki coupling reaction catalyzed using Pd catalyst [290–300]. Saito et al. have reported a nickel (Ni) catalyzed cross-coupling reaction of inert aryl chlorides with boronic acid (Scheme 40) [293,294].

The Suzuki cross-coupling reaction of aryl or vinyl halides was extensively studied, employing Ni and phosphine complexes catalysts systems [295]. Recently, Sharma et al. developed a magnetically recoverable silica-based nanosized hybrid catalyst and utilized it in the Suzuki coupling reaction (Scheme 41) [296]. The target cross-coupling reaction was accomplished under relatively mild conditions and with a high yield. The catalyst was found stable and highly active [296].

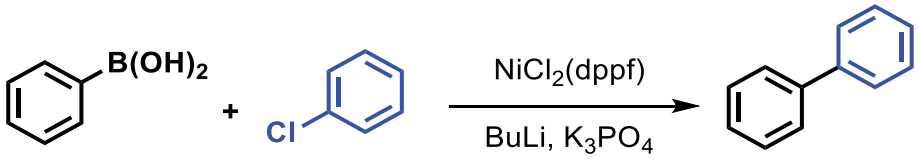
In another study, chitosan was functionalized with an alkylated imino-thiophene ligand through a click reaction route and used to stabilize Ni [297]. This nanocatalyst was applied as a low-cost and active heterogeneous catalyst for the Suzuki coupling reaction



Scheme 38. Pd-based Fe_3O_4 catalyzed coupling reaction of aryl halides with phenylboronic acid.



Scheme 39. ZnO as catalyst supported by hydrazone assisted palladacycle for the Heck coupling reaction.



Scheme 40. Ni Catalyzed Suzuki cross-coupling reaction of aryl chloride.

(Scheme 42). The newly designed catalyst was recyclable for up to 6 runs with no leaching or reduction in catalytic activity [298].

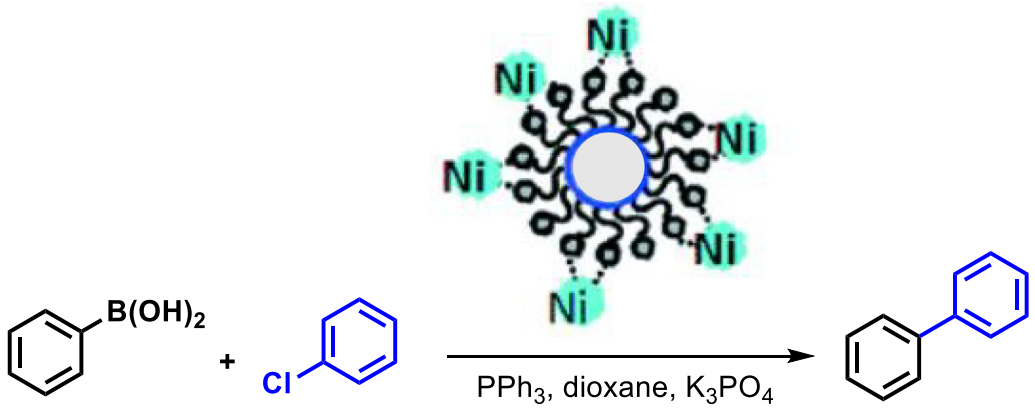
The nickel catalyst has been synthesized by an *in-situ* generation of Ni(0) species because of the presence of glycerol as a reducing agent and then utilized for the Suzuki coupling reaction [299]. This is probably due to a single electron transfer or oxidative addition mechanism [300]. Nagarkar et al. reported the base-free and heterogeneous system for the Suzuki-coupling reaction of the phenylboronic acid with a series of the aryl diazonium salt using glycerol as a reaction medium in the presence of Ni NPs as a catalyst [301]. The Ni NPs-based nanocatalyst was recycled and reused for up to 5 runs. They further reported that DMSO as a co-solvent may enhance the product yield by up to 90 %. Ni nanocatalyst-assisted Heck coupling reaction is still not been investigated like Pd-based catalyst and remains a challenge for future research.

Houdayer et al. have synthesized Ni (0)/polyaniline (PANI) nanocomposite for the Heck coupling reaction between an aryl halide and acrylate/olefine as alkene counterparts [302]. The Ni NPs were easily prepared by reducing $Ni(OAc)_2$ with ^{t}BuONa activated NaH in THF at reflux conditions. The ^{t}BuONa-stabilized Ni (0) particles were then transformed to aniline-stabilized Ni(0) particles by ligand exchange with aniline. The polymerization of aniline-stabilized Ni(0) particles by ammonium persulfate then finally yielded Ni (0)/PANI nanocomposite. The as-prepared nanocatalyst proved to be highly efficient, providing aryl alkynes with a high yield (81 %) (Scheme 43) [302].

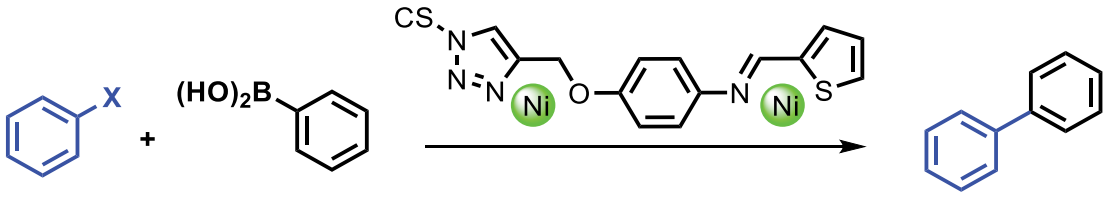
The intermolecular Heck reaction leading to carbocycle formation has been explored by the Alexanian and co-workers [303], the Newman group [304], the Kong group [305,306], and the Jia group [307]. Jarvo and co-workers employed the intermolecular cyclization reaction to introduce the quaternary carbon center into the newly formed carbocycles [308], also with the Garg group [309] and the Desrosier group as well [310]. The utility of nickel catalysis in various Heck coupling reactions is presented in Scheme 44.

Wang et al. employed a potassium fluoride-alumina-supported nanosized nickel(0) powder catalyst for the Sonogashira reaction [202]. Under microwave irradiation and solvent-free conditions, aromatic and aliphatic terminal alkynes were coupled with aromatic-, heteroaromatic-, and vinyl iodides as coupling partners to produce the corresponding products in good to excellent yield (Scheme 45). The alumina acted as a temperature moderator in the microwave reaction. Liquid reactants with metallic nickel without alumina may lead to uncontrollable reactions.

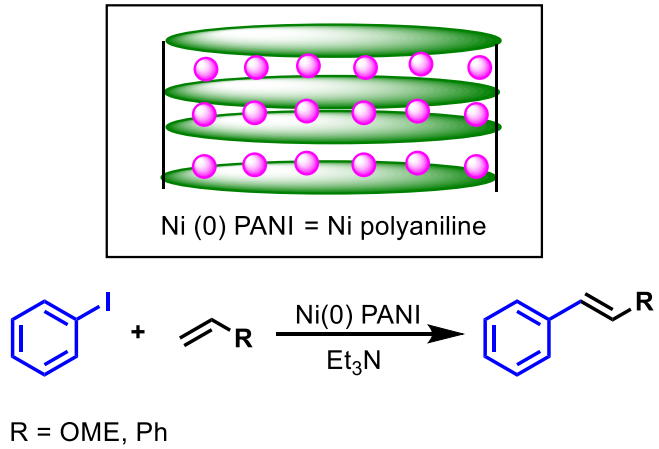
Zhang's group has developed a novel method to produce periodic mesoporous silica (PMO) supported Ni(II) organometallic nanocatalyst [311]. The PMO was functionalized with diphenylphosphine ligands and was then utilized as a support to



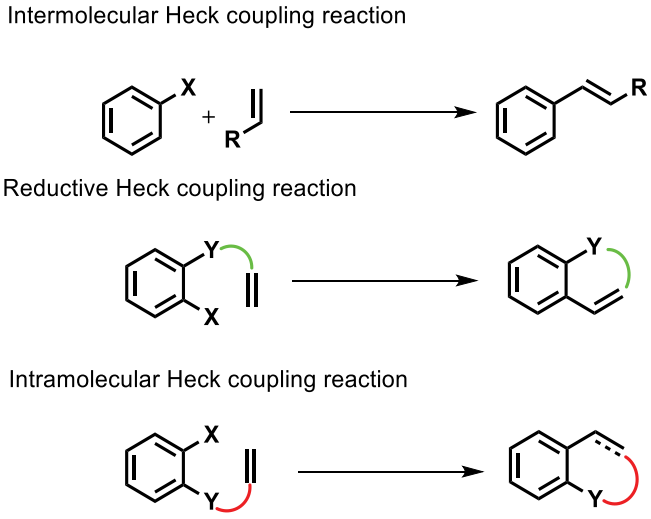
Scheme 41. Ni-based nanocatalyst catalyzed Suzuki coupling reaction [296].



Scheme 42. Chitosan@NP catalyzed Suzuki coupling reaction.



Scheme 43. Ni(0)/polyaniline composites for the Heck coupling reaction.

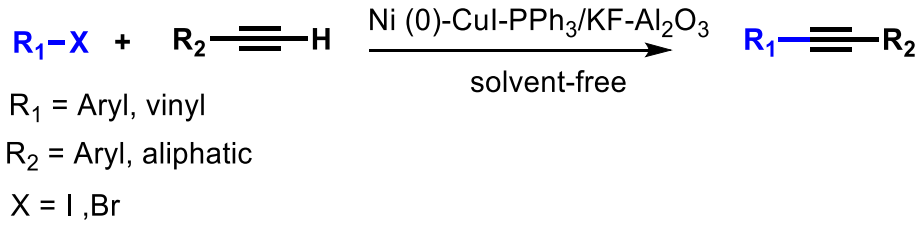


Scheme 44. Application of Nickel catalysis in Heck reaction.

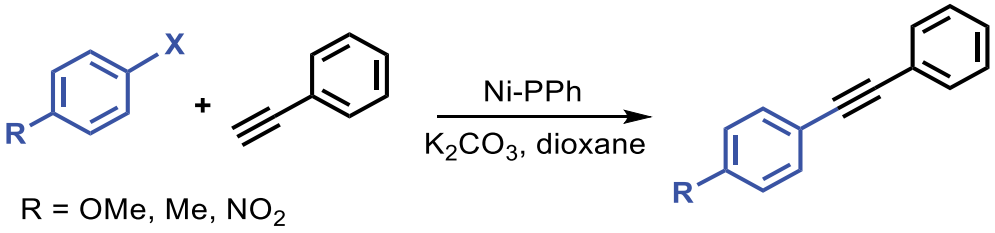
immobilize Ni NPs through coordination interaction. The MPO structure allowed high dispersion of Ni (II) active sites, and the embedded phenyl groups onto the silica walls improved the surface hydrophobicity and promoted the adsorption of reactant molecules onto Ni NPs in an aqueous media. The as-prepared catalyst was employed together with CuI as a co-catalyst in the Sonogashira coupling reaction to produce aryl alkynes in good to excellent yields (Scheme 46). The nanocatalyst exhibited good recyclability, showing no significant activity loss after three repeated cycles.

In another study, Tamimi et al. immobilized the Ni NPs on poly (4-vinyl pyridine)-grafted silica [312]. The deposition of the Ni NPs was confirmed by scanning and transmission electron microscopy. The as-prepared Ni NPs catalyst and CuI as co-catalyst displayed excellent catalytic activity in the Sonogashira cross-coupling reaction under amine and phosphine-free conditions. Both electron-rich aryl iodides and electron-poor aryl bromides sowed excellent reactivity. Nevertheless, less reactive aryl chlorides required the addition of additives such as tetra-*n*-butyl ammonium bromide. Moreover, barely 3 ppm of the NPs leaching was observed after the sixth recyclability runs. Similarly, Das et al. have developed a cost-effective and one-pot approach for synthesizing heterogeneous Ni NPs (1 to 4 nm) supported onto rGO for Sonogashira coupling reaction [313]. The uniform dispersion of Ni NPs onto the surface of rGO nano-sheets was realized without additives or surfactants. The nanocatalyst displayed diverse substrate scope since a series of aryl halides, including heteroaryl halides, electron-rich, and poor aryl halides, were coupled with phenylacetylene to produce the corresponding aryl alkynes in excellent yields (Scheme 47).

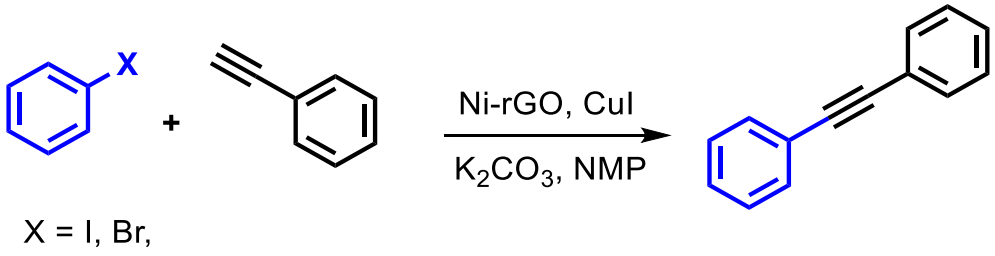
The recyclability of the catalyst was also investigated, wherein the activity decreased from 93 to 68 % after the 6th run. Leaching of Ni NPs was not observed. Likewise, a copper-free Sonogashira



Scheme 45. Ni-powder catalyzed Sonogashira coupling reaction.



Scheme 46. Nano-sized Ni catalyzed Sonogashira reaction in water.



Scheme 47. Ni NPs doped rGO catalyzed Sonogashira coupling reaction of aryl halides and phenylacetylene.

coupling reaction was carried out, employing Ni-ferrite NPs as a catalyst in the presence of water [314]. Overall, the synthetic application of nickel nanocatalysts in Heck reactions is quite prominent in organic synthesis. It reveals excellent interest over the other transition metal catalysts because of its unique characteristic that prepares them to be more significantly applicable in various fields of synthetic applications. Among the transition metals, nickel is chosen for its cost-effective, non-toxicity, and ease of availability. Ni NPs are successfully employed in different catalytic processes, e.g., cycloaddition, cycloisomerization, and multicomponent coupling reactions. In order to further explore the substituent scope of the prepared Ni catalyst for the offered reactions of phenylboronic acid with mesylates, tosylates, and heteroaryl halides. For Suzuki coupling reactions, all the substrates such as mesylates and tosylates, gave a good yield, while heteroaryl halides offered a slightly lower yield [296]. In another important study, the author investigated the effect of various aryl halides having either electron donating or electron withdrawing moieties and also the substituents ortho or para-substituted aryl halides and heteroaryl bromides. Among these substrates, the aryl bromides and aryl iodides converted in a shorter time to their corresponding products with high yields, while aryl chlorides needed more time than the former aryl halides and had lower conversion efficiency. For example, 2-bromoacetophenone instigated its less reactiveness and obtained 54 % conversion efficiency, due to higher steric hindrance of the ortho substituents. Also, the heteroaryl bromides experienced successful couplings to obtain the desired products with high yield and conversion [298]. Hajipour and Abolfathi also demonstrated that the growing steric hindrance extant of ortho-substituted aryl halides offered lower yields than the para-substituted ones [297].

4. Conclusion and future perspective

This article demonstrates that the TM-based NPs catalyzed C—C cross-coupling reactions have witnessed phenomenal growth in the past decade (Fig. 1). The C—C coupling transformation finds its utility in producing biomaterials, fine chemicals, natural products, and medicine [315]. More specifically, the Suzuki, Heck, and Sonogashira reactions have been widely exploited to generate diverse structural frameworks and chemical entities. Since homogeneous Pd catalysts are far from industrial realism, heterogeneous metal nanocatalysts have emerged as a promising alternative owing to their superior catalytic activity, broad substrate scope, and ease of handling. Moreover, the ease of recovery and reusability makes their use a non-hazardous and greener approach, which is important for sustainability and economic viability. For a higher catalytic activity, stability, and recyclability of the heterogeneous metal nanocatalysts system, careful design of the nanocatalysts size, shape, and morphology is important. Likewise, the architectures of well-defined nanostructured supporting materials have also made meaningful advancements in the field. Moreover, the dexterity to integrate the precise nano-sized (1 to 100 nm objects into the composite nanostructured affords unprecedented opportunities to generate multiple catalytic interfaces in a single nanostructured catalyst to exploit the synergistic effects between these catalytic interfaces. Mechanistic understanding of a chemical process is equally important. The organic transformations usually follow multistep complex reaction pathways, which need to be fully understood to understand the role of nanocatalysts. A pedagogical approach is required to empower the synthetic chemist with the suitable software and computational chemistry tools needed to

design their reaction, prediction flowcharts, and probe reaction pathways was envisioned. Computational chemistry tools such as first principle *ab initio* and density functional theory could be a strong means to predict the mechanism and kinetics of the chemical transformation. Computational modeling of nanocatalysis due to the variety of nano sizes of the NPs becomes quite complex. Therefore, the theoretical modeling would be an excellent addition to optimize the large number of trials required to explore various configurations corresponding to the different sizes of NPs [316]. Acquiring 'chemical accuracy' through computational modeling would necessitate considering all aspects of the catalyst environment, such as support, pressure, temperature, solvent, and other surface-bounds species [317]. The atomistic approaches, combined with machine learning tools, can achieve the desired accuracy required to shape optimum nanocatalysts [318].

Similarly, the dynamic model for understanding reaction conditions in a real reaction environment, such as atomic mobility within each NPs can also be employed. Finally, new numerical tools are in high demand to understand the assembly of NPs into super architectures and their influence on catalytic reactions [319,320]. Artificial intelligence technology could be a viable solution to solve grand challenges in organic transformations which need in-depth chemical intuition. Advanced hardware, enhanced algorithms, and growing storage capacity would allow scrutinizing variables beyond the ability of human beings and detection of hidden patterns in insignificant and core scale datasheets. The future of advancement and approaches will ultimately assist the engineering and design of nanocatalysts for the efficient routes (retrosynthesis and prediction of products) with their improvement (optimization of reaction parameters) to identify new chemical matter.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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